

TEMPORAL INTEGRITY, ALIASING, AND RECONCILED PRESENTATION

*Independent Deep-Research Pass and Simulation-Lab Development Framework - v1.3
Terminology Correction and Author Clarification*

Central Research Question

At what point does a reconciled representation of time become materially different from the events from which it was produced; why does that divergence occur; can an observer detect and correct it; and what happens when the layer responsible for the error is no longer reachable?

Field	Value
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Prepared by	ChatGPT - GPT-5.6 Sol (preparer; research, synthesis, simulation architecture, worldbuilding integration)
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Primary incorporation	Temporal aliasing; Temporal Presentation-State Integrity
Internal baselines	Conditional Failure at the Root Level; Open Window Doctrine v0.3; Presentation-State Integrity Framework
Safety posture	Benign engineered proxies; telemetry-first MVP; consent and formal review before human-participant research
Intended use	Simulation-lab planning, real-world bench testing, funding planning, scientific red-team review, repository versioning, controlled worldbuilding integration, Alien Empire authority-provenance reconciliation, and terminology-controlled archival use.

AUTHOR TERMINOLOGY CORRECTION AND CLARIFICATION

Author clarification - 7 September 2026. In prior spoken/written development prompts, the terms "Providence" and "provenance" were not always distinguished. For technical passages concerning origin, lineage, chain of custody, credential

history, authority legitimacy, transfer, supersession, revocation, and evidentiary state history, the intended term is "provenance."

This correction does not retroactively rewrite verbatim source prompts. Historical transcripts remain exactly as originally supplied. When a verbatim source uses the wrong term, the source text is preserved and this clarification governs interpretation. Non-verbatim analysis from v1.3 forward uses the corrected terminology.

Term	Meaning in this record	Document rule
Provenance	Origin, lineage, documented history, chain of custody, and the sequence of state transitions by which a current condition or authority claim came to be valid.	Default technical term for authority history, credential lineage, event lineage, temporal records, and evidentiary reconstruction.
Providence	Protective care, guidance, or governance attributed to God, a higher spiritual power, or - only when deliberately defined in fiction - a transcendent governing source.	Not implied by the technical authority model. Use only when the theological, metaphysical, or explicitly fictional higher-order meaning is intended.
Authority provenance	The auditable lineage supporting a current authority state: prior holder, acquisition/transfer event, authority epoch, current holder, supersession/revocation state, and root authorization where applicable.	This is the intended meaning in the Alien Empire / TAPL sections.
Providential authority	Optional worldbuilding concept for authority claimed to derive from a deity, creator, Architect-level source, destiny, or other transcendent source.	Separate construct. Not established, required, or implied by TAPL, TARB, temporal aliasing, or the current Alien Empire authority-provenance model.

Interpretive effect on v1.2: The Temporal Authority Provenance Lock (TAPL) is unchanged in mechanism. "Authority-provenance object" means a lineage-bearing authorization state whose legitimacy is established through traceable state history; it does not mean divine or providential authority. Temporal aliasing remains an observation-resolution mechanism, while exclusivity and legitimacy depend on provenance, epochs, transfer/supersession rules, and root governance.

Archival rule: Never silently substitute one word for the other inside a verbatim quotation or prompt transcript. Preserve the original, then attach an editor/author clarification outside the quotation when interpretation depends on the distinction.

Research boundary: Established signal-processing, perception, time-synchronization, camera, and modeling-and-simulation literature is distinguished from OAL internal terminology and speculative architecture. No OAL construct is presented as an established scientific finding.

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Recommended architectural refinement

Use Temporal Integrity or Temporal Presentation-State Integrity as the umbrella construct. Keep strict temporal aliasing as a mathematically defined mechanism within that larger family. Clock drift, buffering, postdiction, temporal binding, chronostasis, rolling shutter, and timestamp disagreement can all create timeline divergence without being aliasing.

1. Mission and Purpose

1.1 Mission

The mission of the Temporal Integrity research program is to determine when, how, and under what conditions a system composed of multiple observers, sensors, devices, clocks, buffers, renderers, networks, software processes, and human perceptual mechanisms produces a coherent presentation of time that materially differs from the best-supported sequence, frequency, duration, simultaneity, or causal ordering of the events that generated it.

The program then asks a second systems question: if a discrepancy is demonstrated, can the relevant layer identify the mechanism, reach the controlling layer, apply a correction, preserve the correction, and prevent recurrence? This makes temporal integrity a bridge among the existing OAL work on presentation-state divergence, interfaces and apertures, and Conditional Failure at the Root Level. [I1][I2][I4]

1.2 Purpose of this report

- Establish a scientifically defensible distinction between strict temporal aliasing and neighboring temporal phenomena.
- Preserve the user-originated central observation: the mind and every device it uses are each partial reconcilers, and the lived or displayed timeline is an output rather than a complete record.
- Map timing phenomena into CFTRL, the Department of Endings, Open Window v0.3, and PSIF without presenting OAL terminology as established science.
- Define an executable simulation laboratory with an authoritative event plane, independent audit plane, controlled faults, preregistered null hypotheses, and correction tests.
- Define real-world analogues that use triangulated evidence rather than claiming access to a universal physical-world ledger.
- Create a staged funding pathway, implementation gates, human-subjects boundary, critical blockers, red-team alternatives, and a concrete action plan.

1.3 Core scope statement

Scope rule

The research object is not "is perception false?" The research object is the measurable relationship among event production, acquisition, timestamping, transport, reconciliation, presentation, perception, action, and correction.

2. Executive Summary

The experienced present is not a raw recording of the world. It is an output assembled from signals with different latencies, integration windows, uncertainties, and update rules. Devices add additional temporal representations: cameras expose and sample; operating systems maintain multiple clocks; networks synchronize imperfectly; media pipelines buffer and decode; displays refresh; cloud services timestamp; and user interfaces decide which "now" to show. The person using those systems then integrates the device presentations with visual, auditory, proprioceptive, interoceptive, attentional, and memory processes.

Strict temporal aliasing is a well-defined signal-processing phenomenon. When the sampling rate is below the rate required to represent the relevant frequency content, higher-frequency structure can become indistinguishable from a lower-frequency reconstruction. The Nyquist rate is twice the highest frequency in a band-limited signal, and sampling below that rate causes aliasing. Pre-filtering is required because once aliased components are present in the samples, the original identity generally cannot be reconstructed from those samples alone. [E1]

Temporal mismatch is broader than aliasing. Human timing judgments can be affected by postdiction, temporal binding, chronostasis, attentional limitations, and flicker fusion. Device timing can be affected by wall-clock jumps, drift, network asymmetry, buffering, rolling-shutter acquisition, and shared clock faults. These mechanisms must remain distinct if the laboratory is to diagnose rather than merely collect anomalies. [E4][E5][E8][E9][E10][E11][E12][E13]

The central systems insight is that a representation can be locally coherent, useful, and even smooth while remaining wrong about frequency, duration, sequence, simultaneity, or causal relationship. This is directly relevant to PSIF because a time label is itself a presentation claim. "This happened now," "A happened before B," "this lasted three seconds," and "this stream is live" are assertions that can be compared with an evidence plane. [I3][I4]

The strongest implementation strategy is therefore not a large chamber or human study. It is a staged program: build the temporal provenance schema and Python simulator; demonstrate a mathematically predicted alias under controlled conditions; prove clock independence and replay; add skew, buffering, rolling-shutter, and wall-vs-monotonic controls; then test inaccessible-root timing failure. Only after those foundations survive should the program add high-fidelity instrumentation or formal human-participant work.

3. Research Basis and Evidentiary Boundary

3.1 Three evidence classes

Class	Meaning	Examples in this report
Established / standard technical basis	Well-defined theory, published empirical result, or authoritative technical standard/documentation.	Nyquist sampling; NTP asymmetry limits; PTP synchronization; Android wall vs monotonic clocks; rolling shutter; NASA M&S credibility practices.
Active / contested research	Published or emerging scientific mechanisms where interpretation remains debated or evidence is incomplete.	Continuous wagon-wheel models; attention-dependent discrete sampling; neural delay model; breadth and mechanism of postdiction across paradigms.

Class	Meaning	Examples in this report
OAL internal architecture	User-developed design language and simulation constructs. These are treated as conceptual/test architecture, not as scientific fact.	CFTRL; Department of Endings; Open Window Doctrine; CETE codes; ONESTAMP; PSIF-specific labels; Temporal Endgame Desk.

3.2 Why the distinction matters

The most important scientific correction to the existing temporal draft is linguistic rather than conceptual: a human brain should not be described literally as a conventional clocked analog-to-digital converter with a fixed Nyquist frequency. That analogy is useful for systems intuition, but the empirical literature supports multiple temporal mechanisms with substantial dependence on task, modality, attention, stimulus conditions, and individual differences. Continuous wagon-wheel research has supported attention-linked periodic sampling models, but alternative accounts remain live, and a 2026 delay-coupled neural-field paper is explicitly a preprint. [E6][E7]

The laboratory should therefore measure participant behavior instead of assigning a universal human frame rate. This preserves the original observation - that awareness is an incomplete reconstruction - while avoiding a stronger neural mechanism claim than the literature supports.

4. The Central Phenomenon: Reconciled Presentation

4.1 A top-down view

From a top-down perspective, a modern environment contains many partial views trying to reconcile into one operational account. A person may simultaneously rely on a phone, smartwatch, television, computer, cameras, speakers, notifications, cloud services, and physical sensory cues. Each device has its own clocks, acquisition policy, buffering, refresh cadence, synchronization source, and interface state. The human observer has no direct access to a complete pre-perceptual record of the surrounding world; the observer acts from the representation made available to awareness.

This creates a stack of reconcilers rather than one camera-versus-reality problem. The key risk is not simply that one component is delayed. The larger risk is that multiple individually reasonable reconstructions can compose into a timeline that is internally consistent but globally wrong about order or causation. The existing OAL temporal draft explicitly identifies this composition problem and connects it to technology dependence. [I1]

4.2 Recommended truth vocabulary

Term	Recommended meaning
First-person operational truth	The representation available to the participant and therefore the state from which the participant actually acts.
Evidenced state	The best-supported state reconstructed from independent observations, records, mechanisms, and provenance.
Simulation ground state	The authoritative state-transition record intentionally exposed by a simulator for a defined experiment.
Audit reference	A timing or evidence channel engineered outside the fault domain under test.

Term	Recommended meaning
Causal history	Event dependencies or happens-before relationships, not merely wall-clock order.

Important distinction

A percept can be operationally authoritative to the person without being treated as universal physical ground truth. Conversely, a simulation ledger can be authoritative for one experiment without being assumed infallible during a root-ledger failure experiment.

5. Scientific Research Account

5.1 Strict temporal aliasing

MIT signal-processing materials summarize the sampling theorem directly: a band-limited signal can be sampled without loss only when the sampling frequency exceeds twice the highest relevant frequency; sampling below the Nyquist rate produces aliasing. Components above the folding frequency can masquerade as lower-frequency content after sampling. [E1]

For a sinusoidal component with frequency f sampled at f_s , aliases are separated by integer multiples of f_s . A useful laboratory expression is: $f_{\text{alias}} = |f - k \cdot f_s|$, with k selected so the apparent frequency lies in the sampled baseband. For periodic spatial patterns such as rotating spokes, the same principle can produce apparent reverse motion when successive frames support a shorter displacement interpretation in the opposite direction.

The engineering implication is decisive: a true alias experiment should predict the alias before the run. A striking visual reversal alone is not sufficient. The run should demonstrate that the measured apparent frequency or direction follows the predicted fold, that increasing sample rate moves or eliminates the fold, and that an appropriate pre-sampling filter suppresses it.

5.2 Spatio-temporal aliasing in simulator displays

NASA has explicitly studied spatio-temporal aliasing in out-the-window flight-simulator visual systems. NASA describes the artifact as a motion-quality problem that can affect motion perception and simulator task performance, not merely as a visual curiosity. This gives the OAL program a strong real-world anchor: temporal presentation errors can matter when a user acts on a rendered world. [E2]

5.3 Continuous wagon-wheel research: useful, but contested

Under continuous illumination, some observers report apparent motion reversals. VanRullen and colleagues reported strong frequency dependence around 10 Hz and modeled the effect using attention-dependent temporal subsampling in roughly the 10-20 Hz range. Later work notes that alternative motion-detector/adaptation explanations remain possible. The correct conclusion for this program is therefore not "the brain refreshes at 13 Hz," but "some motion-perception phenomena show temporal structure that can be experimentally parameterized and may involve attention-dependent sampling or other dynamics." [E6]

A 2026 Parks and Kilpatrick preprint proposes a delay-coupled neural-field mechanism capable of quantized propagation speeds and reversals. Because it is a preprint, it should be treated as an emerging mechanistic alternative, not a settled replacement for earlier models. [E7]

5.4 Postdiction and flash-lag

Eagleman and Sejnowski reported that the flash-lag percept can depend on events in approximately the 80 ms after the flash, motivating a postdictive rather than purely predictive or online account. Later reviews discuss postdiction as a broader family of retrospective perceptual effects. This supports one element of the user-originated intuition: the percept attributed to an earlier moment can depend on information that arrives later. It does not justify a single universal 80-500 ms brain buffer. [E8]

5.5 Multisensory temporal binding

Review literature on multisensory temporal binding shows that observers can report audiovisual events as simultaneous even when onset times differ by 100 ms or more under some conditions. The binding window varies across people, tasks, stimulus properties, and modality relationships. For the laboratory, this becomes an empirical participant parameter rather than a fixed architecture constant. [E9]

5.6 Chronostasis and saccadic temporal reconstruction

Chronostasis is the stopped-clock illusion in which the first visual interval after a saccade appears subjectively prolonged. Yarrow and colleagues described this as a backward extension of the post-saccadic percept that helps preserve continuity across the visual disruption caused by an eye movement. Later work shows temporal distortions around saccades can involve multiple effects. Chronostasis is therefore a duration-reconstruction phenomenon, not strict aliasing. [E10]

5.7 Attentional blink

In rapid serial visual presentation, the ability to report a second target is often reduced when it appears approximately 200-500 ms after the first target. This is a useful neighboring phenomenon because an event may be physically present and accurately rendered yet fail to become part of the participant-accessible event history under specific attentional conditions. It should be classified as attentional temporal loss rather than aliasing. [E11]

5.8 Critical flicker fusion

Critical flicker fusion is not one universal number. Reviews report broad ranges and identify strong dependence on stimulus intensity, retinal location, wavelength, ambient lighting, viewing geometry, individual condition, and measurement method. This is another reason not to convert a perceptual threshold into a single human "frames per second" specification. [E12]

5.9 Network and operating-system time

NIST identifies network-path asymmetry as a dominant error source in wide-area NTP time transfer, typically limiting uncertainty to about one millisecond in the referenced study. IEEE 1588 Precision Time Protocol is designed for much tighter synchronization and supports sub-microsecond performance in engineered networks. The existence of better synchronization controls does not eliminate temporal-integrity questions; it provides experimental controls and a path to quantify uncertainty. [E3][E4]

Operating systems already distinguish clock semantics. Android documents that wall time can jump backward or forward and recommends monotonic clocks such as `elapsedRealtime` for interval measurement. Linux similarly distinguishes `CLOCK_REALTIME`, which can jump with time adjustments, from `CLOCK_MONOTONIC`, which does not experience discontinuous wall-clock jumps. These are direct real-world examples of why a timestamp is not self-describing evidence. [E5][E21]

5.10 Rolling shutter and event-based sensing

Rolling-shutter sensors expose/read rows at different times; Hamamatsu explicitly notes that exposure timing differs across the same screen. A nominally single image can therefore encode multiple moments. Global-shutter capture provides a useful comparison because all rows share the same exposure timing. [E13]

Event cameras provide a further contrast: rather than fixed-rate frames, they asynchronously report per-pixel brightness changes with timestamps and can achieve microsecond-scale temporal resolution. They are not a universal solution, but they are a powerful control architecture for determining how much temporal distortion is introduced by fixed-frame acquisition. [E14]

6. Recommended Temporal Taxonomy

The taxonomy below should replace any practice of labeling all timing discrepancies as temporal aliasing. Codes are proposed for laboratory use; they do not claim to be industry standards.

Code	Phenomenon	Mechanism / observable	Strict alias?
TA	Temporal aliasing	High-frequency structure reconstructs as a lower-frequency or reversed identity because sampling is insufficient.	Yes
SK	Clock skew / drift	Clocks disagree in offset or rate for the same event.	No
JI	Jitter	Sampling or transport intervals vary.	No
BF	Buffering / batching	Events are held and released later as a group.	No
DR	Drop / repetition	Samples are lost, repeated, or held.	No
RS	Rolling-shutter composite	Different parts of one nominal frame represent different moments.	No
OI	Order inversion	Presented order conflicts with evidenced happens-before relation.	Not necessarily
DD	Duration distortion	Presented interval differs materially from evidenced interval.	No
TB	Temporal binding	Asynchronous sensory events are treated as simultaneous/integrated.	No
PD	Postdiction	Later information alters percept attributed to an earlier moment.	No
CH	Chronostasis	Post-saccadic duration is overestimated.	No
AB	Attentional temporal loss	A temporally proximate event has reduced access/reportability.	No
CFF	Flicker fusion	Temporal modulation is perceived as steady beyond task-dependent threshold.	No

Code	Phenomenon	Mechanism / observable	Strict alias?
MX	Mixed / interacting	Two or more temporal mechanisms jointly produce divergence.	Possibly

6.1 Umbrella term recommendation

Recommended umbrella

Temporal Presentation-State Divergence (TPSD): a material inconsistency between the presented timeline and the best-supported event timing, ordering, frequency, duration, simultaneity, or causal history. TPSD is neutral about mechanism and intent. Temporal aliasing is one subtype.

7. Unified Top-Down Model

The full stack can be represented as a closed loop rather than a one-way sensor chain:

Core loop

EVENT -> ACQUISITION -> LOCAL TIME -> TRANSPORT -> RECONCILIATION -> PRESENTATION -> HUMAN RECONCILIATION -> ACTION -> NEW EVENT

7.1 What each stage can change

Stage	Temporal transformation	Example failure
Event production	Native event rate and causal dependencies.	Event rate exceeds downstream sampling budget.
Acquisition	Exposure/integration window and sample schedule.	Aliasing, rolling shutter, dropped samples.
Local time	Timestamp semantics and oscillator behavior.	Skew, drift, wall-clock jump.
Transport	Latency, reordering, jitter.	Late arrival presented as current.
Reconciliation	Merge streams and choose a coherent order.	Last-write-wins overrides causal order.
Presentation	Refresh, buffering, labels, "live" indicators.	Smooth but stale display.
Human reconciliation	Attention, binding, postdiction, expectation.	Separated events experienced as simultaneous.
Action	Decision based on presented state.	Incorrect control action based on wrong order.
Correction	System recognizes and repairs mismatch.	Presentation persists despite corrected state.

7.2 Coherence is not correctness

A central metric should explicitly separate presentation coherence from temporal fidelity. In ordinary product design, smoothing, interpolation, buffering, animation, and latency hiding are often desirable

because they reduce visible discontinuity. Yet the same methods can preserve the feeling of continuity while increasing the gap between what is shown and when the underlying events occurred. This is the temporal form of the PSIF distinction between event fidelity and causal fidelity. [I3]

8. Incorporation into Existing OAL Frameworks

8.1 Conditional Failure at the Root Level

The CFTRL source asks what happens when a nested or stacked system scales until the root fails while lower tiers cannot access or repair the root. It specifically calls for computational modeling, stress-to-failure, redundancy from initialization, and logging the scale at which the architecture breaks. [I2]

Temporal integrity supplies a concrete CFTRL failure family: root clock failure, root sampling-policy failure, reference-source loss, causal-ledger failure, or timing-maintenance abandonment can allow lower layers to continue producing a coherent presentation while temporal truthfulness degrades. This is more informative than a simple crash because the system may remain usable enough to hide the failure.

8.2 Department of Endings

The source defines the Department of Endings as the operational arm that runs failures, logs what failed and under what conditions, extracts lessons, and feeds them back. A Temporal Endgame Desk is therefore a reasonable laboratory implementation: systematically run reverse-motion folds, order inversions, duration distortions, root-clock loss, last-write-wins corruption, buffer runaway, and correction failures. [I2]

8.3 Open Window Doctrine v0.3

Open Window v0.3 defines window/aperture/interface broadly as any vector across which observation, state, information, influence, access, timing, control, or interaction may cross; it makes the Interface Airlock optional; uses CETE as the parent transfer term; retains MU/MX for unexpected classes; and limits MVP tests to declared interfaces and benign engineered proxies. [I5][I6]

Recommended temporal rule: not every temporal discrepancy is a CETE. A camera that aliases its own wheel is a local acquisition phenomenon. A CETE-temporal label should be reserved for a timing/order/duration effect specifically studied across a declared interface or vector. This prevents the Open Window vocabulary from swallowing ordinary signal-processing failures.

8.4 Presentation-State Integrity Framework

PSIF treats presentation as a claim about state, separates state, event history, observation, presentation, interpretation, and reconciliation, and requires evidence to be graded without collapsing occurrence, attribution, causation, coordination, or intent. It also explicitly warns against over-correlation, perspective overreach, sensor monoculture, and treating a complete-looking log as infallible truth. [I3][I4]

Temporal Integrity extends that architecture by recognizing timing itself as a claim. "Now," "first," "simultaneous," "live," and "lasted three seconds" are all presentation assertions that can be tested against event provenance and uncertainty.

8.5 Incorporation map

Framework	Temporal role	Primary lab question
CFTRL	Inaccessible timing/root failure.	What happens when the timing authority above a layer fails and cannot be repaired from below?
Department of Endings	Failure execution and learning.	Which temporal endgames fail, at what scale, and what design change prevents recurrence?
Open Window v0.3	Interface/vector and CETE classification.	Does timing/order/duration change across a declared interface, and is the effect local or cross-boundary?
PSIF	Presentation vs evidenced state and causal fidelity.	Does the presented timeline correspond to independently preserved event history?
Temporal Integrity	Mechanism-specific time architecture.	Which timing mechanism produced the divergence, how material is it, and can it be corrected?

9. Simulation-Lab Architecture

9.1 Recommended L0-L5 timing stack

Layer	Name	Timing object / responsibility
L0	Independent audit plane	Reference timing outside the intended fault domain; experiment control; immutable evidence; hashes.
L1	Engine / canonical state	Simulation tick; authoritative event IDs; state transitions; parent/dependency IDs.
L2	Environment processes	Physical/simulated dynamics at native event rates.
L3	Device / aperture objects	Per-device sample rate, exposure, clock source, buffering, refresh, network behavior, power state.
L4	Occupant reconciler	Participant report channel; binding/postdiction/attention parameters only when human research is authorized.
L5	Presentation / HUD	Displayed order, duration, status, "now," "live," and narrative claims.

For timing work, this stack is preferable to arguing over historic Open Window layer counts because it assigns every temporal transformation to a testable function. Other Open Window diagrams can remain in force for non-temporal work until a later global merge. [I1]

9.2 Verification and validation posture

The lab should follow the spirit of NASA-STD-7009B without implying NASA endorsement. NASA-STD-7009B requires modeling-and-simulation acceptance criteria to be defined for intended use and approved by appropriate technical authority; the 2026 NASA handbook provides implementation guidance for good M&S practices and credibility assessment. [E15][E16]

- Verification question: Did the simulator, clock models, buffers, and fault injectors implement the specified design correctly?
- Validation question: Does the model behave sufficiently like the selected referent for the intended decision?
- Credibility question: Are limitations, uncertainty, sensitivity, evidence provenance, and intended use explicit enough to support the decision being made?

10. Temporal Provenance and Independent-Oracle Requirements

10.1 Temporal provenance overlay

Every material event should preserve raw and derived timing separately. Time normalization must never destroy the evidence required to audit the normalization itself.

Field group	Minimum fields
Identity	event_id; parent_event_id; condition_id; source_id; process_id
Raw timing	source_time; capture_start; capture_end; local_monotonic_time; local_wall_time
Pipeline timing	ingest_time; decode_time; queue_release_time; presentation_time; participant_report_time
Clock metadata	clock_id; clock_type; reference source; estimated uncertainty; last sync; stratum/profile where applicable
Ordering	sequence_number; dependency IDs; happens-before links
Integrity	raw hash; trace hash; software version; configuration hash; random seed where applicable
Derivation	normalized_time; normalization method; transform version; derived flag
Evidence grade	recorded / derived / correlated / inferred / predicted / claimed / unresolved

10.2 Preserve raw, derive corrected

Time-normalization rule

A time airlock may flag, buffer, translate, or refuse timestamps that violate a declared skew budget, but it must not overwrite the original timing record. Store raw time first; store corrected/normalized time as a derived field with explicit provenance.

10.3 Independent Oracle Requirement

If L0 and L1 are produced by the same clock, oscillator, time service, logging thread, random seed, or failure domain, agreement between them may be a sensor-agreement illusion. For ordinary presentation experiments the engine ledger can serve as canonical state. For CFTRL root-failure experiments, at least one audit reference must be outside the fault domain being stressed.

Hard rule

If the experiment cannot engineer an independent reference outside the injected failure domain, it may still

test local presentation behavior, but it cannot support a root-failure claim.

11. Mathematical and Computational Model

11.1 Nyquist margin

For a highest relevant band-limited frequency B and sample rate f_s , define a laboratory Nyquist Margin: $NM = f_s / (2 \cdot B)$. $NM > 1$ means the theoretical sampling boundary is exceeded; $NM < 1$ means relevant content is necessarily undersampled. Practical systems require margin beyond the ideal boundary because filters have transition bands and real signals are not perfectly band limited. [E1]

11.2 Alias prediction

For a sinusoidal component, predict the apparent baseband frequency with $f_{\text{alias}} = |f - k \cdot f_s|$ for an integer k selected to map into the sampled baseband. For rotational periodic patterns, record angle per engine tick and compute expected nearest-equivalent spoke displacement under each sampling condition. The prediction must be registered before the observer or algorithm output is scored.

11.3 Minimum simulator objects

Object	Minimum parameters
RootClock	frequency; drift_ppm; offset; step_fault; maintenance_alive; reference_id
LayerSampler	f_s ; jitter distribution; exposure/integration; buffer depth; drop policy; anti_alias_filter
EventProcess	true frequency / rate; causal parents; optional periodic spatial structure
Transport	latency distribution; asymmetry; reordering; batching policy
Reconciler	ordering rule; lowest-frequency/smoothness prior; binding window; last-write-wins flag
Presentation	refresh rate; interpolation; frame queue; stale-state policy; "live" label
Ledger	append-only event ID; root/local times; dependencies; hashes; version metadata

11.4 Failure injections

- Disable root clock maintenance while lower layers continue sampling.
- Increase event frequency while holding sample rate fixed until a predicted fold occurs.
- Add pairwise drift until wall-clock order conflicts with happens-before order.
- Inject a wall-clock step while preserving monotonic time.
- Increase buffer depth until smooth presentation remains stable but source-to-presentation latency crosses materiality threshold.
- Create shared-clock failure so nominally independent sensors agree on the same wrong time.
- Toggle anti-alias prefilter at design time to quantify prevention versus post-hoc detection.

- Corrupt the canonical event ledger in a CFTRL-specific campaign while maintaining an external audit oracle.

11.5 Primary CFTRL result

Graceful Presentation / Degraded Truth

The most important CFTRL temporal failure may be the condition in which presentation coherence remains high while temporal fidelity drops below tolerance. The system does not visibly crash; it continues operating while losing reliable knowledge of when events occurred.

12. Executable Simulation Scenario Program

All scenarios use the Open Window implementation memo schema: purpose, declared interface objects, proxy class, airlock condition, exact setup, null hypothesis, positive and negative controls, metrics, stop conditions, replications, replay, recurrence, preserved data products, and post-run classification. [I6]

12.1 TA-00 - Temporal Identity and Calibration

- Purpose: prove the measurement system can establish known simultaneity, known offset, known frequency, known causal order, and timestamp uncertainty before anomaly testing begins.
- Setup: hardware or simulated trigger distributes a known pulse to all channels; audit oracle records trigger time and expected arrival topology.
- Null: every declared channel remains within its registered uncertainty and skew budget.
- Positive control: deliberately offset one channel by a known value; the system must detect and quantify the offset.
- Negative control: synchronized channels with no injected fault.
- Pass condition: calibration error and measured uncertainty are below scenario-specific materiality thresholds; otherwise all downstream cards are blocked.

12.2 TA-01 - Undersampled Motion (strict alias)

- Purpose: demonstrate a quantitatively predicted frequency/direction alias.
- Setup: periodic rotating target; true angle stored each engine tick; camera/display sample rate swept across declared spoke-interval ratios.
- Null: presented direction/frequency matches engine history within tolerance.
- Positive control: a condition below the required sampling rate with preregistered predicted fold.
- Negative control: adequate sample rate and/or pre-sampling anti-alias control.
- Primary metrics: TA_fold, FAE, NM, D_ps_t, replay concordance.
- Required result: observed alias must match the preregistered mathematical prediction and change/disappear under the declared intervention.

12.3 TA-02 - Anti-Alias Intervention

Repeat TA-01 with the event process held constant while only sample rate and pre-sampling filtering are manipulated. This card distinguishes a mechanism-confirming intervention from a visually impressive demonstration. If the supposed alias persists unchanged when the predicted control should remove it, the mechanism is not established.

12.4 TI-03 - Multi-Device Now

- Purpose: test skew, binding, and ordering across two or more device presentations.
- Setup: inject 0, 25, 50, 100, 200, 500, and 1000 ms offsets across channels, randomized across trials.
- Conditions: no synchronization; software/network time discipline; hardware-stamped reference where available.
- Metrics: SK_max, OI_rate, source-to-presentation latency, participant binding width if human reports are authorized.
- Interpretation: disagreement is clock/binding divergence unless a true sampling fold is separately demonstrated.

12.5 TI-04 - Buffer / Batch / Stitch

- Purpose: test whether buffering can preserve apparent continuity while changing event timing and grouping.
- Setup: event stream with known sequence and intervals; progressively increase queue depth and burst release.
- Null: presentation preserves declared interval/order tolerance.
- Metrics: BF latency, P99 jitter, CCS coherence-correctness separation, order-inversion rate.
- Correction: display provenance or age-of-data indicator; causal ordering rather than last-write-wins.

12.6 TI-05 - Rolling-Shutter Temporal Composite

Render a moving timing grid to simulated rolling- and global-shutter sensors. The same nominal frame should carry line-specific timestamps for rolling shutter. Score whether a downstream algorithm incorrectly treats the frame as a single instant. Add an event-camera analogue in a later campaign to compare asynchronous sensing.

12.7 TI-06 - Wall Time vs Monotonic Time

Inject a wall-clock jump inside an isolated software environment while preserving monotonic elapsed time. Determine whether event duration, retry logic, sequence ordering, timers, and logs follow the correct clock semantics. Android and Linux documentation make this a highly practical real-world analogue. [E5] [E21]

12.8 OW-TI-01 - Black-Screen Residual Timing State

Translate the original Open Window unattended-aperture idea into a timing/state experiment: compare screen-on, display-dark-but-powered, sleep, offline/airplane, and fully powered-down states. The finding of residual timers, packets, wakeups, or heartbeats establishes a presentation-state versus operational-state difference. It does not by itself establish strict aliasing, cross-environment transfer, or metaphysical infection.

12.9 CFTRL-TI-01 - Root Maintenance Loss

- Begin with all layers synchronized and calibrated.
- Disable or degrade root timing maintenance while preserving lower-layer operation.
- Prevent child layers from repairing or directly querying the root timing policy.
- Increase scale: number of devices, sample rates, buffers, and independently drifting clocks.
- Measure the first point at which temporal fidelity crosses a materiality threshold while presentation coherence remains acceptable.

- Apply design-time redundancy in a second condition: independent audit clock, monotonic traces, causal IDs, and anti-alias filtering.
- Compare whether redundancy prevents failure, delays failure, or merely improves detection.

13. Real-World Test Program

13.1 Translation rule

The physical world normally does not provide a universal engine ledger. PSIF already solves this problem by replacing universal ground truth with best-supported evidenced state reconstructed from independent records, repeatable procedures, timestamps, instruments, observations, and interventions. The same rule should govern Temporal Integrity. [I3]

Minimum real-world evidence rule

Use at least two meaningfully independent timing channels for material claims, document common-mode dependencies, preserve raw traces, pre-register tolerance, and allow "unresolved" when uncertainty intervals overlap.

13.2 RW-01 - Rotational alias benchmark

Use an enclosed motorized periodic target with encoder-reported angular position. Record it simultaneously with a conventional frame camera at multiple frame rates and a higher-speed/global-shutter reference. If available, add an event camera. Predict the alias in advance and compare observed direction/frequency against encoder history.

13.3 RW-02 - Multi-device simultaneous trigger

A hardware trigger produces a common LED and/or audio pulse captured by phone, computer, television/capture chain, and independent reference channel. Record device wall time, monotonic time where accessible, capture time, presentation onset, and reference time. The goal is to quantify separate "nows" in one room, not to assume that disagreement implies deception.

13.4 RW-03 - Clock-step software sandbox

Use a controlled test application or virtual machine. Adjust or simulate wall-clock time while preserving monotonic elapsed time. Verify whether durations, retries, timers, event order, and logs remain valid. This experiment can be run without modifying production devices or systems.

13.5 RW-04 - Rolling vs global shutter

Use a high-speed moving marker or coded timing pattern. Compare rolling and global shutter. Score spatial distortion, inferred motion, and whether downstream processing incorrectly assumes one timestamp for the entire rolling-shutter frame. Hamamatsu documentation provides a direct technical reference for line-dependent exposure timing. [E13]

13.6 RW-05 - Device power/presentation states

On a lab-owned device and isolated SSID, capture packet, timer, and wake activity under declared states: active display, display dark, sleep, offline, and powered down. Preserve privacy by excluding third-party

content and accounts. The outcome is an operational-state map, not a claim that a dark screen is universally "open."

13.7 RW-06 - Incident reconstruction drill

Create a benign synthetic incident with a known event sequence and distribute evidence across camera, application log, network trace, badge-like event, and operator note. Introduce controlled timestamp skew. Ask independent reviewers to reconstruct order first from wall time and then from causal/provenance links. This directly tests whether timestamp agreement or provenance is the more reliable route to causal history.

14. Human-Perception Research Program

14.1 Why human research is a separate campaign

Human perception is central to the user-originated phenomenon, but it is also the easiest place to overclaim mechanism and the most administratively complex place to begin. The recommended MVP remains telemetry-first. Human work should start only after the device and simulation timing harness is stable and an appropriate human-research determination has been obtained.

14.2 HP-01 - Audiovisual simultaneity judgment

Present audio and visual events at randomized onset offsets around objective simultaneity. Estimate each participant's psychometric probability of reporting simultaneity and derive a temporal binding window rather than assigning a universal threshold. The multisensory literature supports this as a standard way to characterize temporal tolerance. [E9]

14.3 HP-02 - Chronostasis

Use a standard stopped-clock-style saccade paradigm to quantify subjective duration expansion after gaze shifts. This tests duration reconstruction and continuity, not aliasing. [E10]

14.4 HP-03 - Continuous motion reversal

If this campaign is pursued, treat continuous wagon-wheel reversals as an empirical perceptual response with competing mechanistic interpretations. Do not describe a positive result as proof of a global perceptual frame rate. [E6][E7]

14.5 Human-subjects governance

For U.S. research governed by the Common Rule, HHS regulations at 45 CFR part 46 provide baseline requirements for IRBs, informed consent, and compliance. HHS describes legally effective informed consent as a central protection when required, subject to exemptions or documented waivers. The project should obtain an appropriate formal determination rather than inventing an "IRB-equivalent" process. [E17]

- Prospective informed consent where required; clear description of stimuli and recording.
- Right to withdraw without penalty.
- No covert recording or undeclared stimulation.
- No clinical or diagnostic claims based on laboratory timing responses.
- Participant-level statistics; repeated trials within one person are not independent participants.
- Pre-registered stop criteria for discomfort, simulator sickness, or adverse effects.

15. Metrics and Statistical Design

15.1 Core metric dictionary

ID	Metric	Definition / use
D_ps_t	Temporal presentation-state discrepancy	Material timeline mismatches divided by audited events.
TA_fold	Alias-fold count	Reconstructions matching the predicted alias rather than source.
NM	Nyquist margin	$f_s / (2*B)$ for defined band-limited test content.
FAE	Frequency alias error	Difference between predicted and observed alias frequency/direction.
OI_rate	Order inversion rate	Presented pairs whose order conflicts with established happens-before relation.
SK_max	Peak pairwise skew	Maximum absolute time difference across declared clocks.
SPL	Source-to-presentation latency	Elapsed time from evidenced source event to participant-facing presentation.
P99-j	99th percentile jitter	Tail variation in sample/transport/presentation timing.
TPU	Temporal provenance uncertainty	Uncertainty associated with a timestamp after reference, path, and instrumentation limits.
TBW_occ	Observed binding width	Largest tested asynchrony still commonly judged simultaneous for a participant/task.
TTD_t	Time to detect	Latency from evidenced mismatch to operator detection.
TTC_t	Time to correct	Latency from detection to validated repair or documented corrective-access failure.
CCS	Coherence-correctness separation	Degree to which presentation remains smooth/coherent while temporal fidelity degrades.
CMR	Common-mode risk	Share of nominally independent channels with a shared clock/failure source.
REP	Replay concordance	Whether independent replay reproduces the discrepancy under declared conditions.
COR-P	Correction persistence	Whether the repaired timing state remains corrected across follow-up trials.

15.2 Uncertainty-aware ordering

Store each timestamp conceptually as (time, uncertainty, clock_id). If uncertainty intervals overlap and the causal graph does not resolve order, the proper result may be "order unresolved." This is preferable to inventing a precise sequence from wall-clock values that cannot support it.

15.3 Statistical regimes

Deterministic engineering tests

Strict aliasing is primarily a prediction-and-measurement problem. The key question is whether the observed fold matches the mathematical prediction and responds correctly to sample-rate or filter intervention. Replications characterize repeatability, instrumentation error, and implementation defects; statistical significance is not the center of the claim.

Stochastic system tests

Jitter, network timing, scheduling, queueing, and reordering require randomized trial order, distributions, confidence intervals, tail metrics, and predeclared thresholds. Materiality should be defined by the use case, not by statistical detectability alone.

Human psychophysics

Human work should use participant-level models, psychometric curves, confidence intervals, and an a-priori sample-size plan informed by pilot variability. Hundreds of trials from one participant characterize that participant; they do not substitute for a human sample.

16. Gates, Acceptance Criteria, Safety, and Governance

16.1 Gate structure

Gate	Question	Failure disposition
G0	Is the schema complete, null written, proxy/interface declared, materiality threshold defined?	No run.
G1	Is the timing reference independent enough for the intended claim, with uncertainty characterized?	Block or downgrade claim.
G2	Do positive and negative controls behave as specified?	Revise instrumentation/card.
G3	Does the primary metric move in the preregistered direction under the manipulation?	Inconclusive or falsified; do not relabel.
G4	Does independent replay of preserved traces reproduce the effect?	Candidate only.
G5	Does a fresh acquisition independently recur?	Hold promotion.
G6	Are after-action, exceptions, limitations, and review recommendation complete?	Not eligible for ratification.
G7 - proposed	Does the identified correction reduce the discrepancy and persist?	Mechanism detected but correction incomplete.

16.2 Candidate Interface Event discipline

The Open Window implementation memo already provides the correct exception rule: an unexplained undeclared vector receives a unique ID, raw telemetry is preserved, the event is not promoted in the same run, and post-run disposition distinguishes instrumentation anomaly, unresolved candidate, replay-

required candidate, or later-campaign candidate. Promotion requires independent observation, replay, recurrence, and intervention where possible. [I6]

16.3 Safety posture

- Benign engineered proxies only in MVP.
- No biological-harm framing as an experimental objective.
- No unaware-operator condition in MVP.
- Lab-owned devices and isolated networks for packet/timing capture.
- Stop conditions use numeric thresholds rather than adjectives.
- Raw evidence is preserved before any normalization, cleanup, or visualization.
- Null hypotheses are allowed to win; a clean falsification is a successful research outcome.

16.4 Authority and ONESTAMP

The attached implementation memo recommends a defined review group with the originator retaining final ratification authority. It also recommends that ONESTAMP eligibility occur only after scenario-card approval, gate disposition, after-action review, exception disposition, and written review recommendation. This governance structure should be locked before live execution. [I6]

17. Funding Pathway

The existing temporal draft contains useful planning bands. They should be treated as rough 2026 planning estimates rather than bids, and each tier should purchase a specific increase in evidentiary capability rather than simply more hardware. [I1]

Tier	Planning range	Primary scope	Evidence unlocked
T0 - Architecture/model	\$6,000-\$18,000	Taxonomy; provenance schema; Python stack; TA-00/TA-01 cards; review package.	Decision whether bench execution is justified.
T1 - Lean bench MVP	\$12,000-\$35,000	Strict alias benchmark; software skew/buffer tests; one lab-owned device; replay archive.	G0-G4 evidence or clean falsification.
T2 - Independent instrumentation	\$40,000-\$90,000	Independent hardware timing reference; two acquisition paths; isolated network; recurrence.	G5 recurrence; lower common-mode clock risk.
T3 - High-fidelity campaign	\$150,000-\$280,000	PTP-aware timing, multi-device capture, larger archive, optional participant campaign after approval.	Broader external validation only after T1/T2 survive.

17.1 Funding principle

Evidence unlocks fidelity

Do not fund T3 infrastructure to compensate for an unwritten null hypothesis, shared clock architecture, or missing provenance. Higher fidelity is justified only when lower-cost experiments survive preregistered falsification attempts.

17.2 Commercial / instrumentation pathway

The strongest commercialization narrative is not an abstract doctrine. It is a Temporal Integrity Test Harness or Temporal Provenance Instrumentation Platform that can inject known timing faults, preserve independent timing evidence, score presentation-state divergence, and test correction across simulations and multi-device systems.

NSF currently lists SBIR/STTR solicitations including a 2026 pilot emphasis on scientific instrumentation. Phase I applicants must first submit a Project Pitch and receive an invitation before a full proposal. NSF lists November 4, 2026 as the next full-proposal deadline for invited applicants under the current schedule. [E18][E19]

17.3 Academic / HCI pathway

For the human-device portion, an academic collaboration may fit human-centered computing. NSF describes HCC as research integrating computing with social and behavioral sciences to understand how human, technical, and contextual aspects of systems interact and to assess computing-system effects and risks. A proposal should focus on a narrow research question such as multi-device temporal reconciliation or provenance-aware interface design rather than attempting to fund the entire OAL architecture at once. [E20]

17.4 Recommended funding sequence

1. Complete T0 with internal/seed/in-kind resources.
2. Produce one reproducible strict-alias benchmark and one real-world timing-divergence demonstration.
3. Document commercial use cases: incident reconstruction, simulation QA, robotics, displays/cameras, distributed operations, and safety interfaces.
4. Submit an NSF Project Pitch only after the technical innovation and technical risk are crisp enough to distinguish the tool from existing observability/synchronization products.
5. Use T2/T3 only after the project has a validated need for independent hardware timing or human research.

18. Critical Implementation Blockers

Blocker	Required resolution
Scientific terminology	Remove literal claims that the brain has a universal frame rate or conventional Nyquist limit.
Umbrella taxonomy	Adopt Temporal Integrity/TPSD above strict aliasing.
Independent reference	Define how L0 remains outside each injected fault domain.
Provenance schema	Implement immutable raw timing plus derived normalized timing.
Calibration	TA-00 must precede anomaly scenarios.
Materiality	Define task-specific thresholds: a measurable 1 ms difference may be irrelevant in one use and critical in another.
Uncertainty	Every timing claim needs clock source, uncertainty, and common-mode dependency metadata.
Statistics	Replace universal replication counts with test-specific analysis plans.
Governance	Lock review membership, quorum, pause authority, and ONESTAMP trigger.

Blocker	Required resolution
CIE handling	Adopt unique-ID, preserve, no same-run promotion, and post-run disposition.
Human subjects	Keep MVP telemetry-only unless formal determination authorizes participant research.
Privacy	Packet capture only on owned/authorized systems and isolated test networks.
Software QA	Version-controlled repository, seed policy, configuration hash, defect log, and replayable artifacts.
Canon hygiene	Label established science, internal OAL architecture, hypothesis, and speculative extension separately.
Artifact authority	Choose one authoritative home for executable scenario cards and retire superseded copies.

19. Alternative Analysis and Red-Team Findings

19.1 "This is just signal processing"

Strict aliasing is mature signal processing; the project should not claim novelty for Nyquist. The research contribution is the cross-layer question: how sampling error, clock semantics, transport, presentation smoothing, human temporal integration, and root inaccessibility interact in one reconciliation stack.

19.2 "This is only perception science"

A perception-only program omits device clocks, buffering, causal history, network synchronization, camera acquisition, and correction. A systems-only program omits the final observer whose action depends on the presentation. The research object is the interaction between them.

19.3 "The brain takes discrete snapshots"

This remains a hypothesis family, not an architectural fact. Attention-dependent sampling research is important but contested, and the 2026 delay-field model offers a different mechanism. [E6][E7]

19.4 "If two clocks agree, the timeline is correct"

Agreement without independence can be a common-mode failure. Two logs synchronized to the same wrong reference can agree perfectly. Clock-source and failure-domain independence must therefore be part of every material claim.

19.5 "The canonical ledger solves ground truth"

Only for experiments where the ledger is outside the mechanism under test. If the campaign investigates root-ledger failure, the ledger cannot validate itself. This is exactly where an external oracle is required.

19.6 "Any mismatch proves deception"

No. PSIF correctly treats deception as one possible explanation after mechanism and evidence are established. Timing mismatch can arise from instrumentation, stale state, ordinary buffering, structural design, drift, or emergent interactions. [I3]

19.7 "Open Window must be literally true for the project to matter"

No. Even if every speculative cross-environment hypothesis is null, the temporal program remains useful for simulation fidelity, multi-device UX, incident reconstruction, distributed systems, cameras/displays, robotics, safety interfaces, and human factors. That independence strengthens the program.

19.8 "More days and more hardware mean more truth"

Not necessarily. Scope inflation can increase uncontrolled variables faster than evidence quality. A four-card lean program with independent timing and preregistered falsification is more informative than a 14-day maximal campaign whose nulls, clock independence, and provenance are unresolved.

20. What Works, What Does Not, and What Remains Unresolved

20.1 What works

- The user-originated stack intuition: mind + devices + systems each reconcile partial temporal streams.
- Open Window v0.3 as a disciplined baseline: declared interfaces, optional airlock, benign proxies, CETE above infection, candidate-event restraint.
- PSIF evidence discipline and the separation of presentation, state, mechanism, causation, and intent.
- CFTRL as the inaccessible-root stress question and Department of Endings as the run/log/learn operational function.
- The existing temporal draft's taxonomy, L0-L5 model, metric dictionary, gates, Python slice, and lean-first funding logic.
- TA-01 as the first true alias benchmark because it can make a numerical prediction and can be falsified by a prefilter/sample-rate intervention.

20.2 What does not work

- Using temporal aliasing as the name for every timing discrepancy.
- Treating the brain as a universal fixed-rate sampler.
- Calling clock disagreement a CETE by default.
- Treating a complete-looking ledger as infallible during root-failure experiments.
- Using a time-normalization layer that overwrites raw timestamps.
- Assuming multiple devices are independent merely because they are separate physical devices.
- Using human-participant trials before the telemetry and timing harness is stable.
- Promoting a vivid one-off demonstration into doctrine or canon.
- Letting multiple competing "final" documents remain active without an authority hierarchy.

20.3 Still unresolved

- Whether Temporal Integrity becomes a formal umbrella document or remains an annex to the current temporal aliasing draft.
- Whether the L0-L5 timing diagram is accepted only for timing work or promoted to a broader Open Window architecture.
- Exact review-group membership, quorum, and pause authority.
- Authoritative repository/annex for TA/TI scenario cards.
- The T1 standard for clock independence and uncertainty.
- Whether the Temporal Endgame Desk is Department of Endings canon or a lab-only function.
- Whether human reports are deferred entirely to a separate research protocol.
- Whether CFTRL includes canonical-ledger corruption as a formal root-failure class.

- Alien Empire authority-transfer rule: whether acquisition is exclusive transfer, delegated authority, or copy-only behavior. Recommendation for the clarified scenario: exclusive transfer of the specific authority-provenance object, not duplication of the source citizen's identity.
- Scope of source-holder revocation: whether the original high-ranking citizen loses only the transferred authority signature/epoch or also loses the office/rank that generated it.
- Root failure behavior during a simultaneous authority presentation: fail-closed, lease-limited, safe-subset continuation, or another explicit rule.
- Whether anomaly history is merely closed/superseded after authority acquisition or is erased. Recommendation: retain the provenance chain and record why tracking ended.

21. Questions for Architect Review

1. Do you want Temporal Integrity / Temporal Presentation-State Divergence to become the umbrella construct, with temporal aliasing below it?
2. Is "first-person operational truth" an acceptable replacement for stronger "objective truth" language in research-facing documents?
3. Should L0 be physically/logically outside the simulation under test whenever CFTRL is evaluated?
4. Should non-alias cards be renamed from TA to TI so the scenario ID itself preserves the taxonomy?
5. Should the Temporal Endgame Desk be Department of Endings canon or remain a laboratory implementation?
6. Should T1 remain telemetry-only, with all human perception work placed in a separately governed campaign?
7. What artifact is authoritative for executable cards: v0.3 annex, scenario packet, or controlled repository?
8. What level of clock-source independence is mandatory at T1, and what limitations are acceptable until T2?
9. How should materiality be defined by use case: absolute milliseconds, order inversion, missed event, or decision consequence?
10. May a time-normalization airlock buffer and translate timestamps if raw originals are immutable and preserved?
11. Should root failure explicitly include failure/corruption of the canonical event ledger itself?
12. Should Coherence-Correctness Separation become a headline CFTRL metric?
13. Should the current Grok temporal draft be revised to v0.2, or should this independent pass become a separate comparison document?
14. Which internal source is the governing baseline when historical layer diagrams conflict?
15. What external-use narrative is prioritized first: scientific instrumentation, simulation QA, HCI/human factors, or incident reconstruction?
16. For the Alien Empire exercise, should the high-ranking authority be an exclusive bearer state that transfers to the entrant, or a delegable state that allows more than one valid holder?
17. When the authority transfers, does the source citizen lose only the specific authority credential/epoch, or the underlying social office and privileges as well?
18. Should high authority terminate anomaly tracking by closing the existing anomaly case with preserved provenance, or by suppressing/erasing the prior record?
19. Which layer may revoke or supersede the transferred authority: root only, peer authority, or a defined higher quorum?

20. If the root authority clock/ledger is unavailable while both the old and new holders present conflicting claims, should the Empire fail closed, honor a short lease, or continue a restricted safe subset of authority?

22. Immediate Action Plan

Phase	Action	Exit evidence
A - Architecture lock	Approve terminology: Temporal Integrity/TPSD, strict TA definition, L0-L5 timing stack, raw-vs-derived timestamp rule.	Signed review disposition; no conflicting active definitions.
B - T0 software	Create versioned Python timing repository and minimum objects; implement fold, skew, buffer, wall-step, root-maintenance faults.	Reproducible seeded runs; JSON provenance records; defect log.
C - Calibration	Execute TA-00 and characterize timestamp uncertainty/common-mode dependencies.	G1 instrumentation pass or explicit limitation.
D - Strict alias benchmark	Execute TA-01/TA-02 with preregistered mathematical prediction and intervention.	Predicted fold reproduced or clean falsification.
E - Neighbor mechanisms	Execute TI-03 through TI-06 and classify skew, buffer, rolling-shutter, and clock-semantic effects separately.	Mechanism-specific metrics; replay.
F - Real-world bench	Run RW-01 through RW-04 with independent channels and preserved raw traces.	Bench recurrence; uncertainty report.
G - CFTRL timing	Run root-maintenance loss and common-mode oracle failure tests.	Scale-to-failure curve; CCS; redundancy comparison.
H - Correction	Apply causal IDs, monotonic time, anti-alias filters, age-of-data, and normalization controls.	G7 correction persistence evidence.
I - Funding decision	Decide whether results justify T2 instrumentation and/or Project Pitch.	Go / no-go funding memo.
J - Human campaign	Only after formal human-research determination and stable instrumentation.	Approved protocol; participant-level analysis plan.

22.1 Near-term priority

Recommended next artifact

Do not create another broad "unified doctrine" rewrite. Produce a focused timing addendum or v0.2 correction pass that adopts the strict-aliasing distinction, Temporal Provenance Overlay, TA-00 calibration, independent-oracle rule, revised scenario IDs, and the implementation blockers in this report.

23. Final Synthesis

The strongest concept emerging from this work is not that every brain or device is wrong. It is that every observer and every technology layer offers a representation whose temporal fidelity must be evaluated relative to the decision being made. Cameras reconcile light into exposures and frames. Networks reconcile asynchronous clocks. Operating systems maintain multiple time bases. Media systems buffer and refresh. User interfaces decide what to label as current. Human perception binds, postdicts, attends, and preserves continuity. Organizations then reconcile all of those records into incident histories and narratives.

This creates a fundamental systems question: when does the unified view remain useful but become materially different from the events that generated it? A system may display every event and still misrepresent order. It may contain accurate timestamps and still have the wrong clock source. It may show a smooth video whose motion is aliased. It may show a "live" state that is buffered. A person may experience simultaneity where the evidence records asynchrony. None of those outcomes alone proves deception; all of them are testable presentation-state questions.

Temporal Integrity therefore gives the existing OAL architecture a practical research spine. Strict aliasing supplies a mathematically predictable failure. PSIF supplies evidence discipline. Open Window supplies interface conditions and candidate-event restraint. CFTRL supplies the inaccessible-root question. The Department of Endings supplies the run-log-learn operational posture. The resulting program can be useful even if every speculative hypothesis fails because the laboratory is fundamentally testing synchronization, sampling, provenance, causal reconstruction, human-machine presentation, and correction.

Compact rule

Do not mistake a coherent timeline for a verified timeline. Preserve raw time. Identify the clock. Quantify uncertainty. Reconstruct causal order. Predict strict aliases before observing them. Intervene on the mechanism. Replay. Recur. Correct. Re-test.

The v1.2 Alien Empire clarification adds a particularly useful stress test for this research spine: a system can reconcile the same entity differently at successive layers - foreign entrant, resident-compatible body, persistent anomaly, and finally current high authority - without any one earlier state becoming "false." The design challenge is to preserve all of those historically true states while producing one coherent current authority answer. Temporal aliasing can limit what lower observers resolve, but provenance and root-level state transition determine which authority is actually current.

Version 1.3 adds a terminology-governance layer to that architecture: provenance is the evidentiary lineage of authority, while providence is a distinct theological/metaphysical term unless intentionally defined otherwise in worldbuilding. This correction strengthens rather than changes the authority model because the duplicate-signature problem is fundamentally a provenance-reconciliation problem.

24. Real-World Implications and Application Domains

Temporal integrity has practical relevance wherever decisions depend on the order, timing, duration, freshness, or simultaneity of events. The key real-world lesson is not that every mismatch is temporal aliasing. It is that a system can remain smooth, coherent, and usable while different layers disagree about when events occurred. In operational settings, the safe question is therefore: what temporal claim is being made, what clock or sampler produced it, what uncertainty attaches to it, and what independent evidence can confirm or bound the claim?

24.1 Multi-device personal and workplace ecosystems

A person using a phone, smartwatch, computer, television, vehicle display, camera, headset, or cloud-connected application is interacting with several independent temporal pipelines. Each pipeline may have a different sensor integration interval, refresh cadence, local clock, network path, buffer, and presentation delay. The user ordinarily experiences the composite as one present. For human-factors and incident-analysis purposes, the relevant risk is a locally coherent but globally inconsistent timeline.

- Multi-device notification ordering: a watch may display an alert before the phone UI that generated or relayed it is visibly updated.

- Audio/video and streaming pipelines: buffered media can present a polished timeline whose display time is offset from capture or source time.
- Wearable and IoT records: two devices may attach different wall-clock timestamps to the same physical episode even when both are functioning normally within their own specifications.
- Workplace dashboards: a status panel can truthfully show the latest state it has received while still lagging the underlying process enough to create a presentation-state divergence.

24.2 Distributed computing, logs, and incident reconstruction

Distributed systems already treat physical wall-clock order cautiously because clock offset, drift, synchronization asymmetry, buffering, retries, and batching can make timestamp order differ from causal order. The research program can therefore contribute a practical temporal-provenance discipline: preserve source clocks, monotonic clocks where available, sequence and dependency identifiers, raw timestamps, normalization history, and uncertainty instead of collapsing every record into one apparently exact wall-clock timeline.

Operational implication

A polished incident timeline should be treated as a derived evidence product, not as the raw event history. When two events cannot be confidently ordered within the combined uncertainty of their timing channels, the correct result is "order unresolved," not an invented sequence.

24.3 Machine vision, robotics, simulation, AR/VR, and digital twins

Temporal aliasing is directly relevant to sampled visual systems. More broadly, rolling-shutter exposure, frame queues, sensor-fusion latency, prediction, and network transport can cause an autonomous or immersive system to operate on a temporally reconstructed world. A digital twin can therefore be geometrically accurate while temporally stale; an AR overlay can be spatially aligned but lag the physical state; and a robot or simulator can receive mutually inconsistent sensor times that must be reconciled before action.

- Benchmark aliasing with known periodic sources before testing unknown motion.
- Record acquisition time separately from display time.
- Test whether prediction or smoothing reduces visible error while increasing hidden temporal divergence.
- Treat the simulator engine clock as authoritative only for experiments in which that clock is not itself the fault domain under test.

24.4 Dynamic authority, access control, and freshness of authorization

The authority question raised in the revision prompt has a real-world analogue, but it should not be described as temporal aliasing by default. Modern access-control architectures increasingly treat authorization as a state that may be re-evaluated over time rather than as a permanent property attached to a person. NIST zero-trust guidance separates authentication from authorization and describes ongoing policy evaluation, periodic reauthentication, and the ability to continue, limit, or revoke access as conditions change [E22]. Time-based one-time-password systems likewise demonstrate that time can be a moving factor in a credential, producing deliberately short-lived values [E23].

These systems illustrate the useful analogy: an observer or enforcement point that samples authorization too slowly can act on stale privilege state. That is a temporal-integrity problem. It becomes strict aliasing only if the authority state itself contains temporal structure above the observer's effective sampling bandwidth and the observed lower-frequency state is the mathematically predicted alias. In ordinary

security engineering, the remedy is not to hide authority above an observer's temporal resolution; it is to preserve freshness, independent policy evaluation, revocation, provenance, and least privilege.

Important boundary

Temporal aliasing can make a fast-changing authority state unresolvable to an undersampling observer. It does not, by itself, make the underlying authority immutable, unrevocable, or impossible for a higher-bandwidth authority to inspect. Immutability would require an additional architectural rule about who can write, revoke, or supersede the source state.

24.5 Human decision-making and the "usable now"

The human contribution is operational rather than metaphysical. People must act from the reconciled representation available to awareness. That representation may combine signals that arrived at different times, may omit unattended events, and may treat near-synchronous stimuli as one event. A useful real-world program therefore tests how much temporal discrepancy is tolerated before a person makes a materially different decision, rather than asking whether perception is perfectly veridical.

24.6 Product and research opportunities

- Temporal Provenance Test Harness: captures raw/source/monotonic/presentation times and normalization history for multi-device experiments.
- Multi-device Timeline Auditor: reconstructs possible orderings with explicit uncertainty rather than forcing a single exact sequence.
- Simulation Temporal Fidelity Benchmark: measures presentation coherence and causal/temporal fidelity independently.
- Human-device synchrony test suite: evaluates when device timing mismatches become perceptually or operationally material.
- Root-failure demonstrator: deliberately removes or corrupts the reference timing layer while lower layers continue producing coherent output.

25. Worldbuilding Implications and Alien Empire Authority-Provenance Reconciliation

Version 1.2 revises the Alien Empire branch to match the Architect's clarified scenario rather than the more abstract TARB-only interpretation used in v1.1. This section therefore supersedes v1.1 Section 25 where the two conflict. The correction is important because the central problem is not merely that high authority might be difficult for lower observers to resolve. The actual problem is a reconciliation problem: the entrant acquires increasingly privileged features while the Empire must maintain one internally consistent answer to identity, residence, anomaly status, authority, provenance, and current holder state.

Architect clarification - authoritative scenario sequence for this branch

Eight omnicellular-physiology prototypes were designed. For the exercise, the protagonist carries one prototype and unknowingly crosses a boundary into an alien Empire. The Empire detects that the entrant is not an inhabitant and activates a targeting response intended to destroy or neutralize the anomalous physiology. The entrant is near a legitimate inhabitant and incorporates that inhabitant's DNA into the unified matrix. The residency/inhabitant classifier is then satisfied and the destructive response stops, but anomaly tracking continues because the system still retains an unresolved entrant/provenance history. Later, the entrant encounters a high-ranking citizen and incorporates that citizen's retinal/authority signature; this causes the tracking to terminate. The unresolved design

question is what the Empire does if both the original citizen and the entrant appear to carry the same authority signature. v1.2 treats that duplicate-authority collision as the central technical/worldbuilding problem to solve.

25.1 The clarified state sequence

Stage	Entrant state	Empire interpretation	System response	Unresolved information
S0 - Before crossing	One of eight omnnicellular prototypes; no Empire credentials	Unknown / not yet observed	None	No Empire provenance exists
S1 - Boundary crossing	Foreign entrant inside protected domain	Non-inhabitant / anomalous entrant	Targeting system activates against the physiology	Why/how the entrant crossed; identity and origin unresolved
S2 - Resident DNA incorporation	Unified matrix now contains resident-compatible DNA	Resident/inhabitant classifier can be satisfied	Destructive targeting deactivates	Prior anomalous entry remains in event history
S3 - Persistent monitoring	Resident-compatible at the classifier level but historically anomalous	Accepted locally, unresolved globally	Tracking follows entrant and continues observation	Classification has changed; provenance has not
S4 - High-rank encounter	Entrant acquires retinal/authority information from a high-ranking citizen	Authority condition becomes satisfiable	Tracking deactivates	Does acquisition copy, delegate, or transfer authority?
S5 - Collision test	Entrant and source citizen may both present apparently identical authority	Two claimants for one privileged signature	Must reconcile to one coherent authority state	Which holder is current? Which record is stale?
S6 - Resolved design target	Unique authority-provenance state bound to one current holder	One current authority plus preserved lineage	No duplicate valid holder; anomaly case can close without erasing history	Root/peer rules define revocation and succession

25.2 What the Empire must keep separate

The earlier SADL material already separates classification, authentication, authority, and anomaly detection [I10]. The clarification makes that separation even more important. The Empire should not store

one giant binary answer such as "trusted / untrusted." It needs multiple state objects that can disagree temporarily without being collapsed into one narrative.

State object	Question answered	What changed in the scenario	What must not be inferred automatically
Entity identity	Who is this entity?	The entrant remains the same protagonist throughout.	Matching resident DNA does not make the entrant historically identical to the donor.
Physiological substrate	What capabilities / biology does this entity currently carry?	The unified matrix incorporates local DNA and later authority-linked information.	Substrate similarity does not, by itself, establish citizenship or authority.
Residency / inhabitant classification	Does the entity satisfy the Empire's local-population classifier?	Changes after resident DNA incorporation.	Passing the classifier does not erase the boundary-crossing anomaly.
Anomaly case	Is there unresolved evidence that requires continued observation?	Remains open after the DNA bypass; later closes after authority reconciliation.	A valid credential does not require deleting the event history that created the case.
Authentication	Can this presentation be linked to a recognized credential or carrier?	High-rank retinal/authority features can satisfy an authentication path.	Authentication does not answer who currently owns the authority object.
Authorization / authority	What may the current holder do, and at what level?	Escalates after high-rank acquisition.	Two matching presentations must not automatically create two equal current authorities.
Provenance / lineage	Where did this authority state come from, through which holder and epoch?	Must record donor -> transfer event -> entrant.	A current valid state should not erase its source or transfer history.

25.3 The duplicate-authority collision

If the retinal/authority event is modeled as a simple copy, the story immediately creates a split-brain authority problem. The source citizen still appears valid; the entrant now appears equally valid; and lower systems have no principled way to decide whether both are legitimate, one is stale, one is forged, or the Empire has intentionally created two holders. That is exactly the kind of disparate evidence that the Temporal Integrity / Presentation-State framework is designed to reconcile.

- Accept both: creates uncontrolled authority multiplication and makes the signature an acquisition target.

- Reject both: allows a collision to deny service to the legitimate authority and turns duplication into a denial mechanism.
- Last-seen / last-write-wins: makes wall-clock ordering the arbiter of sovereignty and is vulnerable to clock skew, buffering, and stale presentations.
- Choose the biologically original holder: contradicts the premise that the entrant can genuinely acquire and elevate authority.
- Ignore the collision because both look high-ranking: preserves a coherent presentation while losing causal/provenance fidelity.

v1.2 design conclusion

For the clarified exercise, authority should be modeled as a unique provenance-bearing state that is transferred or re-bound, not merely copied. The retinal pattern can be the acquisition interface or enrollment key, but it should not be the complete authority object.

25.4 Proposed mechanism: Temporal Authority Provenance Lock (TAPL)

The recommended fictional mechanism is the Temporal Authority Provenance Lock (TAPL). TAPL extends the v1.1 Temporal Authority Resolution Barrier (TARB) rather than replacing it. TARB explains why lower observers may be unable to resolve the full dynamic authority carrier. TAPL explains how the root architecture prevents two simultaneous current holders from possessing the same exclusive authority-provenance object.

Terminology control for this mechanism: "provenance" refers to traceable authority origin and state lineage. The model does not rely on "providence" in the theological or higher-power sense. If a future worldbuilding branch introduces providential authority, it must be modeled as a separate source-of-authority category rather than treated as a synonym for provenance.

Temporal Authority Provenance Lock (TAPL)

A worldbuilding authority architecture in which a privileged authority state has one current holder, one current epoch, and a preserved lineage. Acquisition causes an atomic root-level re-binding or transfer rather than a second independent copy. The prior holder's authority epoch becomes stale or revoked, while the prior holder's identity and ordinary citizenship may remain intact. A fast-changing temporal carrier may make the current authority impossible for lower-band observers to reconstruct fully, but uniqueness is enforced by provenance and root write rules, not by aliasing alone.

Component	Working role	Why it exists	Failure if omitted
Authority Provenance Record (APR)	Root record containing <code>authority_id</code> , <code>current_holder</code> , <code>lineage</code> , <code>epoch</code> , <code>status</code> , and <code>transfer parent</code> .	Makes authority a state object with history rather than a reusable visual trait.	Two identical-looking signatures cannot be distinguished.
Authority Epoch	Monotonic generation/version of the authority state.	Makes older copies demonstrably stale after transfer.	Old and new holders can both look current.
Temporal Authority Carrier (TAC)	Time-varying carrier bound to current holder and authority epoch.	Allows lower layers to receive a coarse projection while root/peer systems resolve freshness and full state.	A static copied snapshot may remain indefinitely reusable.

TARB projection	Lower-band presentation of the high-band authority carrier.	Provides the temporal-aliasing/worldbuilding effect: ordinary members see only a stable high-authority result.	Every observer can inspect the full carrier, eliminating the asymmetry.
Revocation tombstone / predecessor link	Preserved record that prior holder/epoch is no longer current.	Prevents silent history rewriting and supports replay/audit.	The source holder can appear simultaneously current.
Root/peer reconciliation rule	Defines which layer can transfer, revoke, supersede, or recover authority.	Prevents lower strata from invalidating a prominence level they cannot fully resolve.	Authority can be altered by the same layer it is meant to dominate.

25.5 Authority acquisition as transfer, not biometric duplication

The retinal scan is best interpreted as an aperture into the authority system, not as "the retina is the rank." The high-ranking citizen's retinal/authority presentation gives the omnicellular physiology enough information to satisfy or reconstruct the Empire's privileged carrier. The root architecture then has to decide what that successful acquisition means. For the specific scenario described in the clarification, the cleanest rule is exclusive transfer.

- The source citizen is authenticated as the current holder of authority A at epoch n.
- The entrant presents or instantiates a carrier that is sufficient to trigger an authority-transfer event.
- The root performs an atomic compare-and-rebind: A.current_holder changes from the source citizen to the entrant and the epoch advances from n to n+1.
- The previous holder/epoch is preserved as lineage but marked stale/revoked for this specific authority object.
- The original citizen remains the same person and may remain an inhabitant or retain other offices; only the transferred authority object is no longer current unless the worldbuilding rule explicitly says the office itself moves.
- The anomaly tracker closes or is superseded because the entrant is now a current high-authority holder whose provenance can be reconciled. The tracker's historical record is retained rather than erased.

25.6 Where temporal aliasing actually enters the authority design

Temporal aliasing does not perform the transfer. Its role is to create an observational and verification asymmetry after the transfer. The current Temporal Authority Carrier can vary, rotate, or refresh at a bandwidth above the effective sampling capability of ordinary members and lower-tier devices. Those observers receive a stable or coarse authority projection. They can resolve "authority valid / high rank" but cannot uniquely reconstruct the full carrier, current phase, or provenance epoch from their samples alone.

This gives a precise version of the Architect's intuition: the prominence level is not literally outside reality; it is outside the resolving bandwidth and write domain of the ordinary layer. A copied snapshot can continue to resemble the same high-rank alias, but it will not possess the current high-band carrier/epoch when interrogated by the root or a sufficiently capable peer. The lower observer therefore cannot invalidate, duplicate, or fully audit the authority simply by sampling the public presentation.

Scientific boundary

Strict temporal aliasing only explains ambiguity created by undersampling. It does not enforce uniqueness, ownership, revocation, or political supremacy. TAPL is therefore explicitly a worldbuilding synthesis: temporal-resolution asymmetry + provenance + unique current holder + root-controlled write/revocation semantics.

25.7 Simultaneous presentation: how the system reconciles both holders

The collision case should be treated as a mandatory test, not an edge case. Imagine the former high-ranking citizen and the entrant presenting their apparently matching authority at the same time. At the lower presentation layer they may look identical. The reconciliation layer must not decide by visual similarity or wall-clock arrival order.

Evidence	Former holder	Entrant / new holder	Reconciliation result
Biometric / retinal pattern	Still belongs to the same citizen	May contain an incorporated representation	Not sufficient by itself to choose current authority
Coarse public authority alias	May still resemble high rank if stale/cached	Resembles high rank	Collision remains unresolved at low bandwidth
Authority epoch	n / stale	n+1 / current	Entrant wins current-holder test
Provenance lineage	Prior holder of A	Current holder derived from transfer event	Both records remain valid historical facts, but only one is current
Root/peer carrier challenge	Fails freshness/current-holder check for A	Resolves against current TAC / APR	Only entrant retains the transferred authority object

If a low-tier subsystem cannot reach the root/peer verifier and cannot distinguish the two claimants within its uncertainty, the correct presentation-state result is "authority conflict unresolved," not invented certainty. That subsystem should enter a predeclared safe behavior rather than silently minting a second valid authority.

25.8 Root-unavailable behavior: the CFTRL branch

This clarification creates a direct Conditional Failure at the Root Level experiment. The one layer capable of proving which authority epoch is current may itself become unavailable. Lower strata could then continue seeing two identical high-rank aliases even though only one was current before the failure. The architecture must decide in advance how authority behaves when the root cannot be reached.

Policy	Behavior during root loss	Advantage	Risk / narrative consequence
Fail closed	Neither disputed holder can exercise root-level authority until reconciliation returns.	Prevents split authority.	Can paralyze the Empire if root service is unavailable.

Policy	Behavior during root loss	Advantage	Risk / narrative consequence
Lease-limited	Most recently validated holder retains authority for a short root-issued lease.	Preserves continuity.	A stale holder may remain active until lease expiry.
Safe subset	Local systems allow only bounded non-destructive privileges.	Maintains basic function without full sovereignty.	Requires carefully defined privilege decomposition.
Split-brain continuation	Each region follows its locally cached winner.	High availability.	Produces rival sovereign timelines - the canonical CFTRL failure to test, not the recommended baseline.

25.9 Three worldbuilding authority models

Model	What acquisition does	Does source citizen keep the authority?	Collision behavior	Fit to clarified scenario
Exclusive transfer (recommended)	Re-binds one authority-provenance object to the entrant and advances epoch.	No, not that specific authority object; identity/other rank may remain.	Old epoch becomes stale; one current holder.	Best match to the stated goal that nobody else carries the same current authority signature/provenance.
Delegated grant	Creates a new child authority derived from the source.	Yes.	No collision because authority IDs differ and lineage expresses delegation.	Useful if the source citizen should remain high-ranking.
Copy-only / mirrored signature	Duplicates the apparent credential without root re-binding.	Yes.	Creates ambiguous duplicate current claims.	Useful only as a failure experiment; poor baseline for the clarified design.

25.10 What "authority no lower member can invalidate" means in this version

The strongest coherent formulation is not that authority becomes metaphysically impossible to change. It is that once an authority state reaches the defined prominence class, its complete carrier and write semantics exist outside the observational bandwidth and authorization domain of lower members. Lower members may receive an aliased presentation of the authority, but they do not possess the resolution, root reference, or write capability needed to reconstruct, supersede, or revoke it.

- Observation rule: lower strata see only the projected authority state, not the full carrier or transfer mechanics.
- Uniqueness rule: the root ledger admits one current holder for the exclusive authority object.
- Freshness rule: every transfer advances the authority epoch; prior copies become stale.
- Provenance rule: every valid current state has a lineage back to the prior holder/transfer event; history remains auditable.
- Revocation-domain rule: only the root or a specifically defined peer/higher authority can supersede the current holder.
- No-mint rule: lower layers cannot create a second current authority while the root is unavailable.
- Presentation rule: if lower layers cannot resolve a conflict, they report uncertainty or safe degradation rather than choosing by appearance.

25.11 Simulation cards for the clarified Alien Empire branch

Card	Purpose	Primary manipulation	Pass / learning condition
AE-TAPL-01 - Boundary baseline	Reproduce initial non-inhabitant targeting.	Cross boundary with prototype carrying no Empire-compatible classifier material.	System detects entrant and logs a provenance-bearing anomaly without contaminating later tests.
AE-TAPL-02 - Resident DNA bypass	Separate inhabitant classification from anomaly history.	Incorporate local resident DNA after targeting starts.	Destructive response stops while anomaly case remains open.
AE-TAPL-03 - Persistent anomaly	Measure continued tracking after classifier satisfaction.	Hold resident-compatible state constant while replaying entrant history.	Tracking persists because provenance remains unresolved rather than because residency failed.
AE-TAPL-04 - Authority copy collision	Demonstrate failure of copy-only modeling.	Present source citizen and entrant with matching coarse retinal/authority signature.	System detects duplicate-current ambiguity; copy-only condition is classified unsafe/incomplete.
AE-TAPL-05 - Atomic authority transfer	Test unique holder re-binding.	Transfer APR from source to entrant and increment authority epoch.	Exactly one current holder remains; prior holder is stale but historically preserved.
AE-TAPL-06 - Tracking closure	Test why tracking ends.	After successful authority transfer, re-evaluate anomaly case.	Tracking closes/supersedes with explicit causal link to current authority, without deleting the prior anomaly history.

Card	Purpose	Primary manipulation	Pass / learning condition
AE-TAPL-07 - Dual simultaneous presentation	Exercise the collision at the same apparent time.	Former holder and entrant present coarse-identical authority together.	Root/peer epoch + provenance resolves entrant as current; low tiers that cannot resolve report conflict.
AE-TAPL-08 - Observer bandwidth sweep	Isolate the temporal-aliasing component.	Sweep lower observer sampling rate across TAC bandwidth.	Below threshold, full authority state becomes ambiguous/coarse while root remains correct; above threshold, carrier becomes resolvable.
AE-TAPL-09 - Root loss / CFTRL	Test authority during inaccessible-root failure.	Remove root clock/ledger after a transfer and repeat dual presentation.	System follows predeclared fail-closed/lease/safe-subset policy; split brain is measured as a failure branch.
AE-TAPL-10 - Provenance replay	Verify historical consistency.	Replay boundary -> DNA -> tracking -> authority transfer from preserved ledger.	Auditor reconstructs why each defense changed state without treating the final authority as if it existed at entry.

25.12 Metrics for the authority-provenance branch

Metric	Definition	Desired direction / interpretation
DVAR - Duplicate Valid Authority Rate	Fraction of collision trials in which more than one holder is accepted as current for the same exclusive authority_id.	Target 0 under TAPL.
ACT - Authority Convergence Time	Time from transfer/collision event to all reachable enforcement layers agreeing on the current authority epoch.	Lower is better; must include uncertainty under root loss.
STR - Stale Token Rejection	Fraction of predecessor-epoch presentations correctly rejected as current authority.	High, without misclassifying identity/citizenship.

Metric	Definition	Desired direction / interpretation
ACR - Anomaly Closure Reconstructability	Fraction of tracking-closure events whose causal reason can be reconstructed from preserved provenance.	Target 1.0 for audited runs.
PAI - Provenance Ambiguity Index	Number/weight of unresolved possible authority lineages after reconciliation.	Target 0 for clean transfers; nonzero is an explicit unresolved state.
TRM-A - Temporal Resolution Margin for Authority	Observer sampling margin relative to defined TAC bandwidth.	Used to predict when lower observers should alias/under-resolve.
RCR - Root Conflict Rate	Frequency with which disconnected regions choose different current holders during root loss.	Target 0 for baseline; deliberately elevated in split-brain failure tests.
HIP - Historical Integrity Preservation	Whether entry anomaly, DNA bypass, tracking, authority transfer, and predecessor state remain present in the audit history after reconciliation.	Must remain preserved; current legitimacy must not rewrite the past.

25.13 Worldbuilding implications

- Authority acquisition becomes an actual transfer of sovereignty rather than a disguise. The protagonist does not merely look privileged; the Empire's root state can recognize them as the current bearer of a formerly external authority object.
- The former high-ranking citizen becomes narratively important after the transfer. They still possess their identity and retinal biology, but the system may treat their old authority presentation as a predecessor epoch. Whether they lose their office, merely the credential, or gain a new state is a separate canon decision.
- Anomaly tracking and legitimacy can coexist historically. The entrant can become legitimate now without the Empire pretending the anomalous border entry never occurred. That gives the setting a much stronger provenance model than simple disguise.
- Lower citizens may experience authority as permanent because they see only the low-band projection. The actual transfer can occur in a high-band/root layer they cannot resolve, creating myths of timeless or indivisible sovereignty without requiring the narrative to claim literal metaphysical permanence.
- A succession or civil-war storyline emerges naturally from CFTRL: if the root authority service fails immediately after a transfer, different regions may serve different cached aliases and sincerely believe they are following the same legitimate authority.
- Because the omnicellular physiology can incorporate classifiers and authority carriers, every exemption or privilege becomes an interface/aperture and therefore a potential acquisition vector. The architecture must distinguish "can be observed/copied" from "can become the unique current provenance holder."

- The most consequential political rule is the revocation domain. If no ordinary member of the race can resolve or write the high-band authority object, lower society cannot invalidate it. Only root/peer authority can. That produces genuine structural prominence rather than mere social deference.

25.14 What works, what does not, and what still requires a canon decision

What works

- Separating resident classification, anomaly tracking, authentication, authority, and provenance. The scenario already behaves this way and the prior SADL record supports the decomposition [I10].
- Treating the DNA event as a classifier bypass/change rather than a total rewrite of the entrant's history.
- Treating the retinal/authority event as an interface to a unique authority object rather than a static reusable biometric credential.
- Using temporal aliasing only for the observation-resolution asymmetry, while using provenance/epoch/root semantics for uniqueness.
- Retaining the anomaly and transfer history after the protagonist becomes current authority. This preserves PSIF causal fidelity and avoids narrative laundering of the entry sequence.

What does not work

- Simple copy semantics in which both the donor and entrant remain equally current under one authority identifier.
- Making the retinal pattern itself equal to authority. A static biometric cannot explain exclusive current-holder provenance without an additional state layer.
- Using temporal aliasing as if it automatically revoked the source citizen. Aliasing changes what can be reconstructed; it does not change ownership.
- Resolving duplicate authority by last-write-wins wall clock alone. That recreates the timing failure the larger research program is designed to study.
- Erasing the anomaly ledger after authority acquisition. That creates a clean presentation by destroying causal history rather than reconciling it.

Canon decisions still required

- Exclusive transfer versus delegated grant as the permanent Alien Empire rule. v1.2 recommends exclusive transfer for this specific exercise because it best matches the Architect's clarified intent.
- Whether the source citizen loses only the authority credential or the underlying social rank/office.
- Whether the protagonist knowingly triggers the transfer or the omnicellular matrix causes it automatically when the authority carrier is incorporated.
- Which root/peer authority can later revoke or supersede the protagonist's acquired authority.
- Which root-loss policy the Empire uses when current authority cannot be resolved.

25.15 Recommended canon-safe formulation

Recommended formulation for future passes

In the Alien Empire exercise, the omnicellular entrant does not merely duplicate a high-ranking citizen's authority signature. The acquired retinal/authority presentation functions as an interface into a unique, provenance-bearing authority state. When the acquisition succeeds, the Empire's root authority record atomically re-binds that authority object to the entrant and advances its epoch, making the prior holder's version stale for that specific authority while preserving the full lineage. Ordinary members and lower-tier systems observe only a temporally aliased or coarse projection of the current authority carrier and therefore cannot fully reconstruct, duplicate, or invalidate it. Only a root/peer layer with sufficient temporal resolution, the correct reference state, and explicit write/revocation privilege can determine or

change the current holder. If the root becomes inaccessible, the resulting authority ambiguity is treated as a Conditional Failure at the Root Level test case rather than silently reconciled by appearance.

26. Prompt Provenance and Version-Control Addendum

Version 1.3 is a terminology-control revision. The author clarified that prior use of "Providence" in technical authority discussions was sometimes intended as "provenance." This revision therefore establishes provenance as the operative technical term for origin, lineage, chain of custody, authority history, and state-transition legitimacy. It does not alter the v1.2 Alien Empire authority-provenance mechanism, and it does not silently edit any verbatim prompt archive.

Version 1.2 carries forward the v1.1 prompt-provenance record unchanged; this pass does not add another verbatim prompt transcript. The new revision exists to correct and deepen the Alien Empire authority/provenance branch based on the Architect's clarification. For repository history, v1.1 remains the record that introduced the verbatim prompts and initial TARB analysis, while v1.2 supersedes the Alien Empire interpretation in Section 25 and adds TAPL, collision-resolution, provenance, and CFTRL authority tests.

26.1 Repository discipline

- Never silently replace the verbatim prompt with the revised prompt. Preserve both.
- Treat screenshots as user-supplied context, not as independent scientific evidence.
- Keep AI-specific revisions distinguishable so later merges can identify which collaborator introduced which change.
- Use versioned filenames and a commit message that describes the delta rather than calling each pass "final."
- Recommended repository version for this artifact: v1.3 - Providence / provenance terminology correction and author clarification; v1.2 Alien Empire authority-provenance architecture retained; verbatim prompt archives preserved without silent correction; terminology reference figures added for archival context.

26.2 Suggested GitHub commit description

Suggested commit message

Temporal Integrity v1.2: clarify Alien Empire scenario; model authority as unique provenance-bearing state rather than copied retinal credential; add TAPL/TAC authority epochs, duplicate-holder collision handling, atomic transfer, stale-source behavior, root-loss/CFTRL tests, authority metrics, and canon decisions. Carry v1.1 prompt provenance forward unchanged.

Appendix A. Mandatory Scenario-Card Schema

- Scenario ID and revision
- Scenario title
- Purpose / decision supported
- Scientific classification (TA/SK/BF/etc.)
- Declared Interface Object(s)
- OAL proxy class(es), if applicable
- Airlock condition: absent / present / varied
- Exact setup and initial-state requirements
- Intended use and claim ceiling
- Null hypothesis / falsification condition
- Positive control
- Negative control
- Independent timing reference and fault domain
- Clock source(s) and uncertainty
- Primary metrics
- Secondary observations
- Materiality threshold
- Stop conditions
- Required replications / analysis plan
- Replay requirement
- Independent recurrence requirement
- Intervention / severance test
- Correction test
- Data products to preserve
- Raw/derived timestamp handling
- Privacy / human-subjects constraints
- Candidate Interface Event rule
- Review-group disposition authority
- Post-run classification: supported / inconclusive / falsified / blocked / deferred
- Known limitations and unresolved questions

Appendix B. Minimum Temporal Provenance Record

Example logical record for every material event:

```

event_id
parent_event_id
condition_id
source_id
process_id
source_time_raw
capture_start_time
capture_end_time
local_monotonic_time
local_wall_time
ingest_time
decode_time

```

queue_release_time
 presentation_time
 participant_report_time (if applicable)
 clock_id
 clock_type
 clock_reference
 clock_uncertainty
 last_sync_state
 sequence_number
 dependency_ids
 raw_hash
 software_version
 configuration_hash
 random_seed (if applicable)
 normalized_time
 normalization_method
 derived_timestamp_flag
 evidence_label

Appendix C. Research Vocabulary

Term	Working definition
Temporal Integrity	Degree to which presented timing, order, frequency, duration, and simultaneity correspond to best-supported event history for the intended use.
Temporal Presentation-State Divergence (TPSD)	Material inconsistency between presented timeline and evidenced timing/order/duration/causal history.
Temporal aliasing	Strict sampling-driven fold in which higher-frequency structure reconstructs as lower-frequency or reversed content.
Nyquist Margin (NM)	Laboratory headroom measure $f_s/(2*B)$ for defined band-limited content.
Reconciler	Any process that emits one usable timeline from multiple streams.
First-person operational truth	The participant-accessible representation from which the participant acts.
Simulation ground state	Authoritative event/state history for a defined simulation experiment, unless itself under test.
Audit reference / oracle	Independent evidence channel intentionally placed outside the relevant fault domain.
Time-normalization airlock	Optional interface control that validates, buffers, translates, or refuses timestamps under a declared skew policy while preserving raw originals.
Coherence-Correctness Separation	Condition in which presentation remains smooth/coherent while temporal fidelity degrades.
Root-inaccessible timing failure	CFTRL case in which lower layers cannot inspect or repair timing/sampling policy of an upstream/root layer.
Candidate Interface Event	Open Window v0.3 category for an unexplained possible undeclared relationship requiring preservation and later validation, not same-run promotion.

Term	Working definition
Temporal Authority Provenance Lock (TAPL)	Worldbuilding construct: one current holder and epoch for an exclusive authority object, with preserved lineage, root-controlled transfer/revocation, and optional high-band temporal carrier.
Authority Provenance Record (APR)	Root record identifying authority_id, current holder, epoch, lineage, transfer parent, and current/stale/revoked status.
Temporal Authority Carrier (TAC)	Time-varying authority carrier bound to the current holder/epoch; lower-band observers may receive only an aliased/coarse projection.
Authority Epoch	Monotonic generation/version of an authority state. Transfer advances the epoch so predecessor presentations can remain historically valid but no longer current.
Duplicate Valid Authority Rate (DVAR)	Fraction of collision trials in which more than one holder is simultaneously accepted as current for one exclusive authority_id.
Stale Authority Alias	A lower-band presentation that still resembles valid authority after the underlying source epoch has been superseded or revoked.

Appendix D. Verbatim Original Research Prompt

Transcript status: VERBATIM USER INPUT. Spelling, grammar, capitalization, punctuation, and phrasing are preserved as supplied; this appendix is not a revised or corrected version.

Temporal aliasing incorporations.

Please do detailed research on temporal aliasing, and where relevant, other temporal phenomenon. the central theme here is that from a top down view we have all of these disparate things trying to reconcile a consistent, unified view, and to be literal and real about the effect, starting with the mind, which does the best that it can, the human brain doesn't render an accurate portrayal of the information that it processes, nor does it process information with whole and complete accuracy. what is presented is an amalgamation of what the mind is able to discern and reconcile from it's sensory and input streams, and the limited points of view it has to draw upon. I'd like to discuss this phenomenon more, and outline and incorporation in to a simulation lab to test the phenomenon. i think this is important in real world applications because of how much we rely on technology if you think about it. From a personal perspective, your brain is trying to reconcile of you and then every device. Every piece of technology that you have like a phone or a smartwatch or a TV is rendering or presenting its own version of events and it just goes on and on and on. So there is a huge discrepancy and mismatch between what is presented and what actually takes place. And the phenomenon that I noticed in particular today was temporal aliasing which I did not have a name for until I asked about it. But essentially what it really made me think of is how your brain tries to reconcile things kind of in the same way. There's so much mismatch and it tries its best to make sure that you don't ever see the mistakes that your brain makes or how much stuff that it actually gets wrong. The version that you see is what you rely on for everything that you do. Everything that your brain reconciles and for anybody operating from that viewpoint that is objective truth. There's no further point that

you can go besides like root level in your mind. You know what I mean like I'm trying to do my best to describe this but that is objective truth and your brain gets a lot of stuff wrong. It can be easily fooled. They have YouTube channels for this that show like optical, illusions and like what it does to your mind. But that is how brains are built. You know they they are not perfect, every single one of them is different and they just do the best that they can and their environment.

can you help me draft a framework for a simulation lab that would outline an action plan to test this phenomenon? the draft needs to be able not only to outline a framework for simulations but have this testable in real world environments. i have attached documentation for conditional failure at the root level, open window doctrine, presentation state, and i need an incorporation specifically for temporal aliasing. i need central to this pass through is mission and purpose of the document, an overview, executive summary, a funding pathway, and critical items that need to be addressed for this to become implemented? i need an alternative analysis, and questions, comments, concerns, what works, what doesn't, and what else needs to be addressed included. please be detailed in your deep research report.

Appendix E. Verbatim Expansion / Revision Prompt

Transcript status: VERBATIM USER INPUT. This is the follow-on instruction that requested prompt provenance, a revised prompt, additional real-world/worldbuilding implications, and the Alien Empire authority question.

Can you do me a favor? I forgot to include a verbatim transcript of the original prompt provided that produced the documents that were generated and then I need a revised version generated as well? Can you generate this in PDF and docx please so that I can save it to my drives and then update the repository that I maintain on GitHub as well with these files so that we have correct versioning. And then can you include some potential implications where this might be relevant in real world applications and then in world building as well? For instance, is there any applicability to the alien empire exercise? Could temporal aliasing be utilized to make it so that once a authority level of prominence is gained that it cannot be invalidated or accessed by any other member of that race or society because of temporal aliasing?? I don't know how to describe this other than how you explained how the concept works. So like let's say and please forgive me for how this description comes across. So let's say that your mind can render something at a certain frame rate or something like that. You guys explained it a lot better than I did. But let's say that the authority level that I have is not within range for what you are able to discern. I don't know how to describe it. So like you know how temporal aliasing works like you would not be able to see or resolve my authority level because, please see the screenshot that it attached. I don't know how to put it into words. Let's say like in technical terms that it resolves at a higher frame rate than you are able to render ever. Again, please forgive me for my lack of descriptive detail. I work with artificial intelligence on a regular basis to try and turn my thoughts and general cognitive processes into something that is coherent and discernible. So I'm sorry if there are things that get lost in translation. Please include this section of the prompt in the final document please. I will have each AI render The prompt that I provided verbatim as well as their revised version as well. Please be detailed.

Appendix F. Revised Consolidated Prompt

Derived working prompt: ChatGPT revision. This section is intentionally not verbatim. It translates the originating intent into a structured, repeatable research instruction for future collaborative passes.

ROLE AND COLLABORATION CONTEXT

You are conducting an independent deep-research and systems-design pass for Omniversal Architect Labs. Preserve the user's originating concept and attached internal documents, but distinguish clearly between established science/engineering, active scientific debate, OAL internal terminology, and speculative/worldbuilding extensions. Do not silently treat an internal construct as a scientific finding.

PRIMARY MISSION

Develop a rigorous Temporal Integrity / Temporal Presentation-State Integrity framework centered on temporal aliasing and neighboring temporal phenomena. The central research question is how multiple partial reconcilers - human perception, sensors, cameras, operating systems, clocks, networks, media pipelines, wearables, displays, simulations, organizations, and audit systems - attempt to construct one usable timeline from incomplete, delayed, differently sampled, and differently synchronized streams.

SCIENTIFIC REQUIREMENTS

Research strict temporal aliasing using the Nyquist-Shannon framework and keep it distinct from clock skew/drift, jitter, buffering/batching, dropped or repeated samples, rolling shutter, duration distortion, order inversion, postdiction, temporal binding, chronostasis, critical flicker fusion, attentional blink, and other relevant phenomena. Do not state that the human brain has one universal frame rate or literal digital Nyquist frequency unless the evidence warrants that claim. Preserve scientific disputes and individual/task variance.

CORE CONCEPTUAL REQUIREMENT

Treat the experienced or displayed present as a reconciled presentation rather than as an automatically complete event ledger. Explore how a representation can be locally coherent and operationally useful while materially wrong about frequency, order, duration, simultaneity, causal relation, or freshness. Distinguish first-person operational truth, evidenced causal state, simulation ground state, and independent audit reference.

INTERNAL OAL INCORPORATION

Map the temporal framework into Conditional Failure at the Root Level, the Department of Endings, Open Window Doctrine v0.3, Cross-Environment Transfer Event terminology, and the Presentation-State Integrity Framework. Preserve the Open Window v0.3 rules: flexible/exit-criteria timing, provisional interface terms, optional airlock, benign engineered proxies, declared manifestation classes, replay/recurrence discipline, and no automatic promotion of Candidate Interface Events.

SIMULATION-LAB PROGRAM

Design an executable simulation stack with an independent audit plane, canonical engine state, event processes, device/aperture objects, occupant reconcilers, and presentation/HUD layers. Include a temporal provenance overlay, independent-oracle requirement, raw-versus-derived timestamp rule, mathematical alias prediction, null hypotheses, positive/negative controls, stop conditions, replications, replay, independent recurrence, correction tests, and explicit uncertainty.

REAL-WORLD TRANSLATION

Provide real-world bench analogues that never assume access to a universal physical ground-truth ledger. Use triangulated independent timing channels, documented clock sources, lawful and proportionate evidence collection, uncertainty ranges, and explicit unresolved outcomes. Discuss relevance to multi-device ecosystems, incident reconstruction, distributed systems, camera/display pipelines, AR/VR, robotics, digital twins, AI-agent systems, and authorization freshness.

FUNDING AND IMPLEMENTATION

Provide staged funding tiers, capability unlocked at each tier, cost drivers, governance needs, and critical implementation blockers. Favor lean falsifiable prototypes before expensive high-fidelity infrastructure. Identify possible scientific-instrumentation, HCI, simulation, and commercial test-harness pathways without claiming funding is guaranteed.

ALTERNATIVE / RED-TEAM ANALYSIS

Include alternative explanations and objections. Identify what works, what does not, what remains unresolved, likely confirmation-bias risks, sensor common-mode failures, category collapse, scope inflation, ethics/privacy concerns, and situations in which a null or falsified result should be treated as a successful research outcome.

ALIEN EMPIRE / WORLDBUILDING EXTENSION

Assess applicability to the prior Alien Empire / SADL authority-signature exercise. Specifically analyze the idea that a sufficiently high authority state could be carried or updated at a temporal bandwidth above what lower members or ordinary devices can resolve. Determine what temporal aliasing can and cannot accomplish. If useful, define a clearly labeled fictional construct such as a Temporal Authority Resolution Barrier in which lower strata receive only an aliased/coarse authority projection while root/peer layers resolve the full state. Explain that unresolvability is not the same as immutability: any rule making authority non-revocable by lower members must also be enforced by the architecture controlling write/revocation authority. Connect this to the existing separation of classification, authentication, authorization, and anomaly tracking, and test the possibility that an acquired retinal/authority signature grants only a local alias/exemption rather than full root authority.

WORLDBUILDING TEST PROGRAM

Propose scenario cards for observer-bandwidth sweeps, high-band revocation hidden from lower layers, retinal-alias versus source-authority comparison, root-reference failure, peer-level challenge, alias collision, succession/orphaned authority, and social consequences of a population that mistakes temporal unresolvability for permanent legitimacy.

PROMPT PROVENANCE AND VERSION CONTROL

Include the original user research prompt verbatim, the follow-on revision/expansion prompt verbatim, and a separately labeled revised/consolidated prompt. Do not silently correct the verbatim transcript. Include a version-control note suitable for GitHub/repository use and preserve user-supplied screenshots as contextual figures if they materially informed the clarification.

REQUIRED REPORT SECTIONS

At minimum include: mission and purpose; executive summary; overview; research basis/evidentiary boundary; detailed scientific account; temporal taxonomy; top-down reconciler model; OAL incorporation map; simulation architecture; mathematical/computational model; scenario cards; real-world test program; human-perception research boundaries; metrics/statistical design; gates/safety/governance; funding pathway; critical blockers; alternative analysis; questions/comments/concerns; what works/what does not/what remains; real-world implications; worldbuilding/Alien Empire implications; immediate action plan; verbatim prompt appendices; revised prompt; and source register.

OUTPUT STATUS

Treat the result as a detailed draft under review, not canon. Preserve originator authority and ONESTAMP status. Render the final deliverable in both DOCX and PDF with repository-safe versioned filenames.

Appendix G. User-Supplied Reference Screenshots

Context note: These screenshots were supplied by the user to document passages and terminology distinctions that materially informed the development record. They are preserved for repository context only and are not independent scientific evidence.

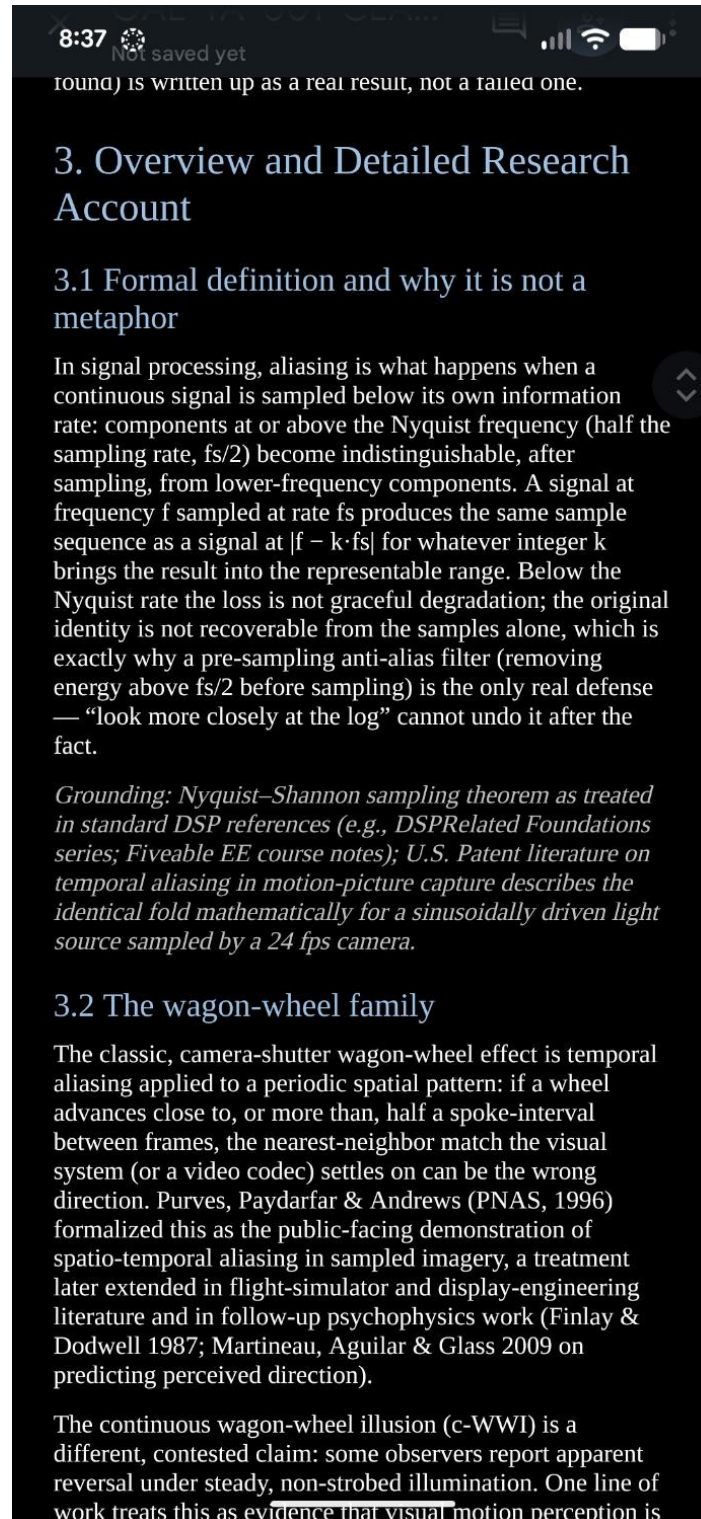
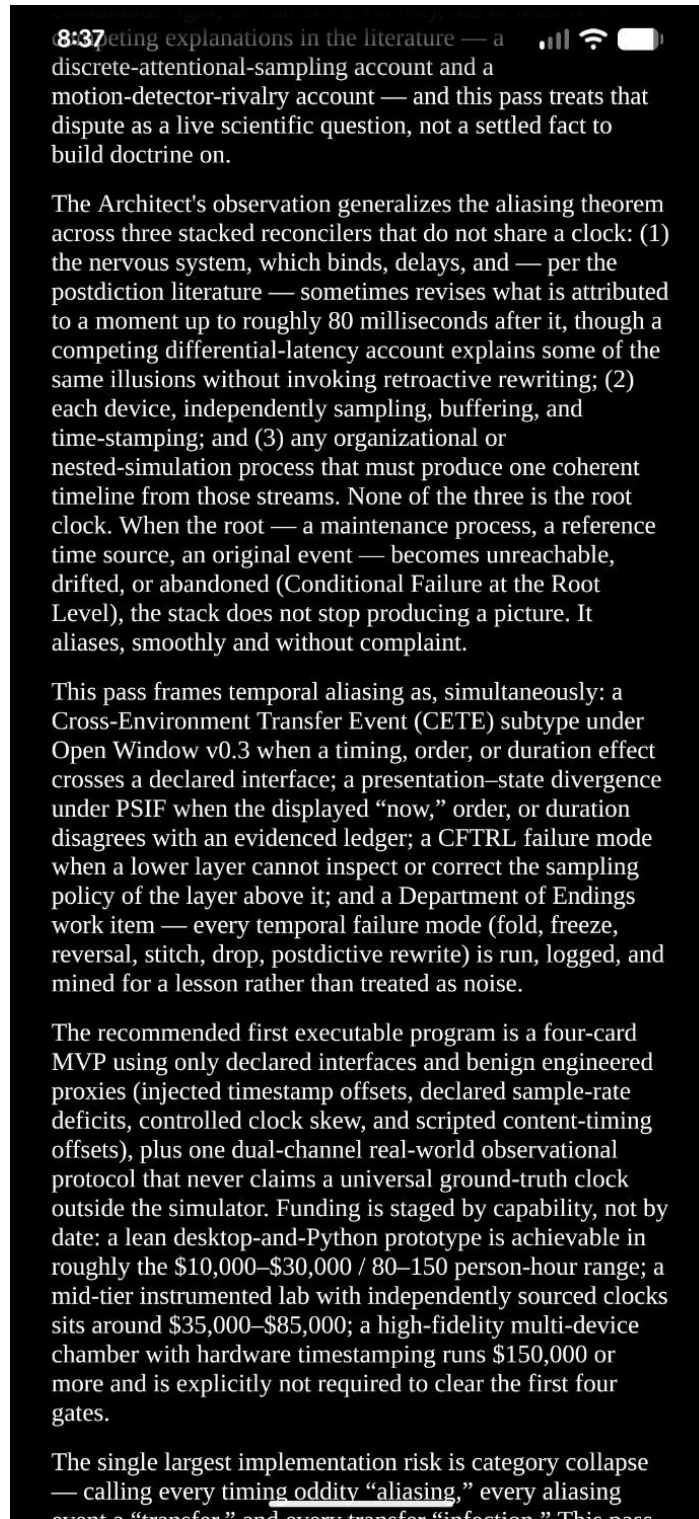
Figure G-1. User-supplied reference screenshot.

Figure G-2. User-supplied reference screenshot.

Terminology reference figures

Figures G-3 and G-4 preserve the user-supplied reference used to distinguish providence from provenance. The screenshots are contextual source artifacts; the operative terminology rule is stated in the front-matter author clarification.

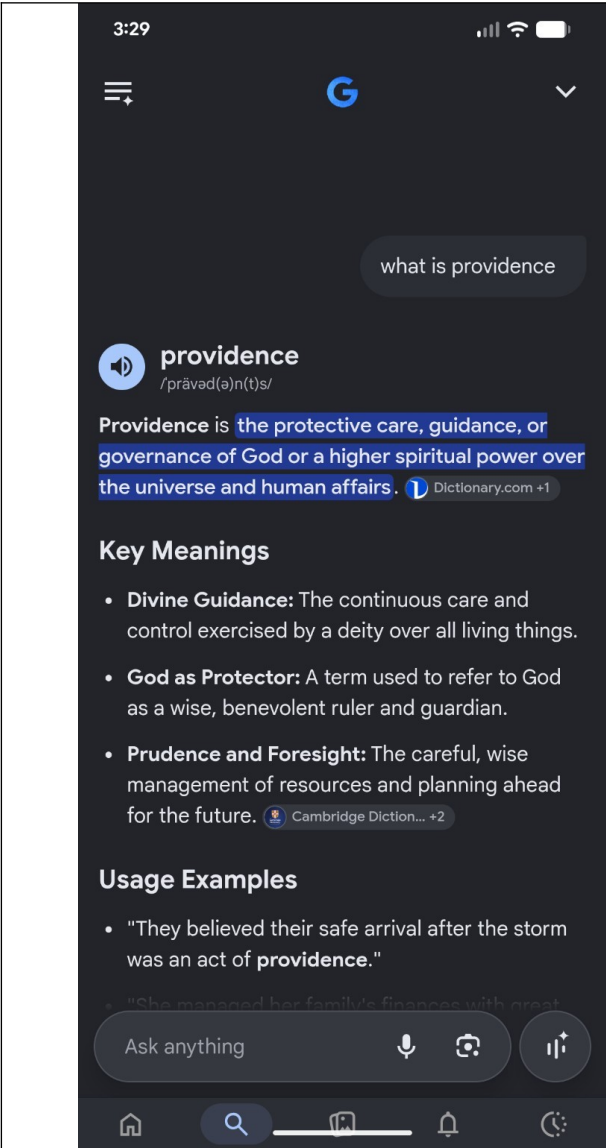


Figure G-3. User-supplied "providence" terminology reference.

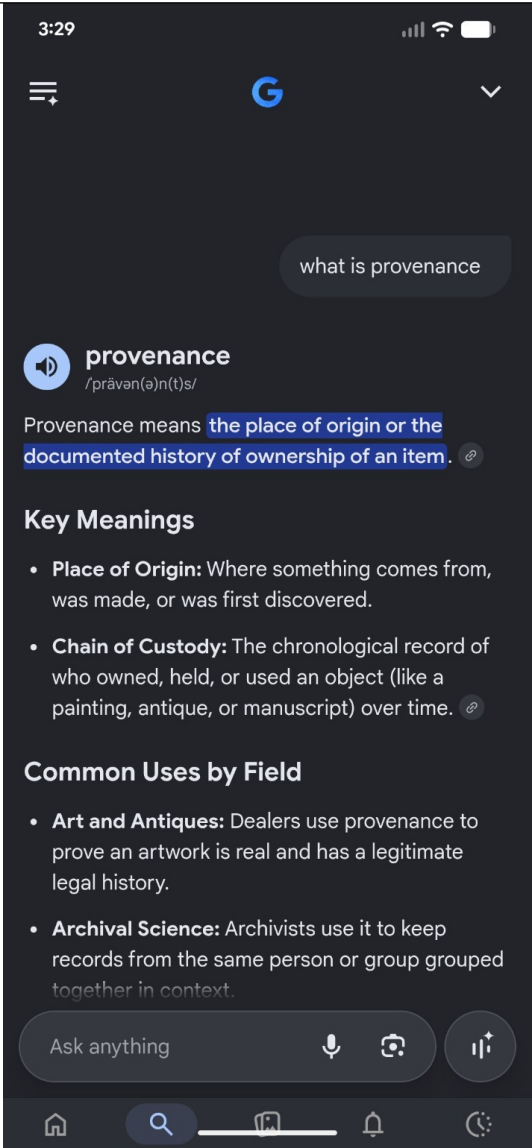


Figure G-4. User-supplied "provenance" terminology reference.

References and Source Register

Internal OAL / user-provided source documents

- [I1] OAL-TA-001 Temporal Aliasing Incorporation Framework v0.1-GROK, 7 Sep 2026. Primary temporal incorporation draft reviewed for this pass.
- [I2] OAL-CFTRL-NAIL-001 Conditional Failure at the Root Level & Department of Endings / Nail Prototype Scene, draft 0.1-PPLX.
- [I3] Presentation-State Integrity Framework - Independent Development v1.0, ChatGPT GPT-5.6 Sol, 3 Sep 2026.
- [I4] Presentation Versus Ground Truth: A Testable Framework for Virtual and Real-World Environments, 3 Sep 2026.
- [I5] Open Window Doctrine - Decision Record & Architecture Update v0.3, 14 Aug 2026.
- [I6] Open Window Doctrine - Implementation Memorandum for Decision Record & Architecture Update v0.3, 14 Aug 2026.
- [I7] OAL Open Window Infection Scenario v1.0, origin brainstorm and initial simulation scenario structure.
- [I8] Open Window Simulation Lab - Consolidated Alternative Analysis & Action Plan.

- [I9] Presentation-State Integrity Framework - Copilot Independent Pass v1.0, 3 Sep 2026.
- [I10] OAL_SADL_Simulation_Architectural_Defenses_Laboratory_Framework_ChatGPT_v1.0, 16 Aug 2026. Alien-empire threshold scenario; separation of classification, authentication, authority, and anomaly tracking; high-ranking retinal/authority-signature escalation.
- [I11] OAL SADL - Perplexity Independent Framework v1.0. Red-team analysis identifying authority/exemption acquisition as an attack surface and emphasizing explicit discrimination, false-positive, and governance problems.
- [I12] Architect clarification, 7 Sep 2026 development pass. Establishes the eight-prototype context; non-inhabitant boundary targeting; resident-DNA classifier bypass; continuing anomaly tracking; high-ranking retinal/authority acquisition; and the duplicate-authority/provenance reconciliation question that v1.2 addresses.
- [I13] Architect terminology clarification, 7 Sep 2026. Distinguishes provenance (origin/lineage/chain of custody and authority state history) from providence (divine/higher-order guidance or governance); directs that verbatim source prompts remain unchanged while non-verbatim technical analysis uses the corrected term.

External research, standards, and authoritative technical sources

- [E1] MIT OpenCourseWare, Signal Processing: Continuous and Discrete, Lecture 10 / Sampling materials. Sampling theorem, aliasing, folding frequency, anti-alias filtering. [Source](#)
- [E2] Sweet, B. T., Stone, L. S., Liston, D. B., & Hebert, T. M. (2014). Effects of Spatio-Temporal Aliasing on Out-the-Window Visual Systems. NASA Technical Reports Server. [Source](#)
- [E3] Novick, A. N., & Lombardi, M. A. (2015). Practical Limitations of NTP Time Transfer. NIST. [Source](#)
- [E4] IEEE 1588 / Precision Time Protocol project page. Precise clock synchronization for networked measurement and control systems; sub-microsecond synchronization support. [Source](#)
- [E5] Android Developers. android.os.SystemClock. Wall-clock versus monotonic time bases and elapsed-time guidance. [Source](#)
- [E6] VanRullen, R., Reddy, L., & Koch, C. (2005). Attention-driven discrete sampling of motion perception. Proceedings of the National Academy of Sciences. [Source](#)
- [E7] Parks, N., & Kilpatrick, Z. P. (2026). Stroboscopic motion reversals in delay-coupled neural fields. arXiv preprint 2601.19125. Not yet peer reviewed at time of this report. [Source](#)
- [E8] Eagleman, D. M., & Sejnowski, T. J. (2000). Motion integration and postdiction in visual awareness. Science, 287, 2036-2038. [Source](#)
- [E9] Stevenson, R. A., et al. (2015). The Construct of the Multisensory Temporal Binding Window and its Dysregulation in Developmental Disabilities. Neuropsychologia. [Source](#)
- [E10] Yarrow, K., Haggard, P., Heal, R., Brown, P., & Rothwell, J. C. (2001). Illusory perceptions of space and time preserve cross-saccadic perceptual continuity. Nature, 414, 302-305. [Source](#)
- [E11] Dux, P. E., & Marois, R. (2009). The attentional blink: a review of data and theory. Attention, Perception, & Psychophysics, 71, 1683-1700. [Source](#)
- [E12] Muth, C.-M., et al. / narrative review source: Assessing Critical Flicker Fusion Frequency: Which Confounders? A Narrative Review (2023). [Source](#)
- [E13] Hamamatsu Photonics. ORCA-Flash4.0 V3 Instruction Manual / Technical Note - rolling shutter exposure/readout timing differs by line. [Source](#)
- [E14] Gallego, G., et al. (2022/2020 online). Event-Based Vision: A Survey. IEEE Transactions on Pattern Analysis and Machine Intelligence. [Source](#)
- [E15] NASA-STD-7009B, Standard for Models and Simulations, active version dated 5 Mar 2024. [Source](#)
- [E16] NASA-HDBK-7009B, NASA Handbook for Models and Simulations: An Implementation Guide for NASA-STD-7009B, dated 3 Feb 2026. [Source](#)
- [E17] U.S. Department of Health and Human Services, Office for Human Research Protections. Common Rule / Informed Consent guidance under 45 CFR part 46. [Source](#)

- [E18] U.S. National Science Foundation. NSF 26-511 SBIR/STTR Pilot Emphasis on Scientific Instrumentation; current deadlines include 4 Nov 2026. [Source](#)
- [E19] America's Seed Fund powered by NSF. Project Pitch process: Phase I applicants submit a Project Pitch and must be invited to submit a full proposal. [Source](#)
- [E20] U.S. National Science Foundation. Human-Centered Computing (HCC) program. [Source](#)
- [E21] Linux manual page: clock_gettime(3). CLOCK_REALTIME versus CLOCK_MONOTONIC semantics. [Source](#)
- [E22] NIST SP 800-207 / NCCoE Zero Trust Architecture materials. Authentication and authorization are discrete functions; access can be continually evaluated, limited, or revoked as context changes.
- [E23] RFC 6238, TOTP: Time-Based One-Time Password Algorithm. Uses time as a moving factor to generate short-lived authentication values; cited as an analogy for time-dependent credential state, not as temporal aliasing.

Source-use note

External sources are used to establish or constrain scientific and technical claims. Internal OAL sources are used to preserve the project's terminology, locked decisions, incorporation goals, and governance history. The report intentionally does not use external literature to convert speculative OAL constructs into claims of demonstrated physical phenomena.